

Optimal Cardinality Constrained Portfolio Selection

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One long-standing challenge in both the optimization and investment communities is to devise an efficient algorithm to select a small number of assets from an asset pool such that a portfolio objective is optimized. This cardinality constrained investment situation naturally arises due to the presence of various forms of market friction, such as transaction costs and management fees, or even due to the consideration of mental cost. Unfortunately, the combinatorial nature of such a portfolio selection problem formulation makes the exact solution process NP-hard in general. We focus in this paper on the cardinality constrained mean-variance portfolio selection problem. Instead of tailoring such a difficult problem into the general solution framework of mixed-integer programming formulation, we explore the special structures and rich geometric properties behind the mathematical formulation. Applying the Lagrangian relaxation to the primal problem results in a pure cardinality constrained portfolio selection problem, which possesses a symmetric property, and to which geometric approaches can be developed. Different from the existing literature that has primarily focused on some direct relaxations of the cardinality constraint, we consider modifying the objective function to some separable relaxations, which are immune to the hard cardinality constraint. More specifically, we develop efficient lower bounding schemes by using the circumscribed box, the circumscribed ball, and the circumscribed axis-aligned ellipsoid to approximate the objective contour of the problem. In particular, all these cardinality constrained relaxation problems can be solved analytically. Furthermore, we derive efficient polynomial-time algorithms for the corresponding dual search problems. Most promisingly, the lower bounding scheme using the circumscribed axis-aligned ellipsoid leads to a semidefinite programming (SDP) formulation and offers a sharp bound and high-quality feasible solution. By integrating these lower bounding schemes into a branch-and-bound algorithm (BnB), our solution scheme outperforms CPLEX significantly in identifying the exact optimal portfolio.

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1. Introduction

The mean-variance formulation pioneered by Markowitz (1952) more than half a century ago laid the foundation of modern financial analysis. Although the variance might not be a perfect measure for the risk (see Markowitz 1959), the mean-variance formulation captures the essence of two conflicting aspects in portfolio management, the risk and the return. In the last few decades, the mean-variance portfolio selection model has been extended and refined in various directions. For example, Li and Ng (2000), Zhou and Li (2000), and Chiu and Li (2006) extend the static mean-variance analysis to the dynamic settings. Considering the uncertainty of the expected return, Costa and Paiva (2002) and Goldfarb and Iyengar (2003) study the mean-variance portfolio selection problem in the context of robustness. Recently, DeMiguel and Nogales (2009) integrate the robust estimation approach into the mean-variance model to overcome an instability of the portfolio weights. Considering some real trading constraints, Li et al. (2006) propose an exact solution approach for the mean-variance model with some discrete features, such as cardinality

constraint, round lot requirement, and concave transaction cost. Bonami and Lejeune (2009) consider the probability constraint in the mean-variance model together with the buy-in threshold, round lot, and diversification constraints. A survey on recent development of mean-variance models can be found in Bonami and Lejeune (2009).

According to Markowitz's classical theory, investors who primarily care about the mean and variance of the portfolio return should construct their optimal mean-variance efficient portfolios from *all* risky assets available in the market. Such an ideal solution in a frictionless world, however, is unattainable in most real-life situations. Due to various forms of market friction, almost all investors invest in only a limited number of assets. Such a gap between the ideal and real worlds motivates us to study the cardinality constrained mean-variance (CCMV) portfolio selection problem, i.e., given an asset pool with total n assets, choose $s < n$ assets optimally from the asset pool to form a mean-variance efficient portfolio. In particular, we focus on the general CCMV portfolio selection problem when short-selling is either allowed or prohibited. Note

that whether short-selling should be restricted has been quite controversial for academics, market regulators, and politicians. Thus, it is meaningful to study both situations with shorting allowed and shorting prohibited. On the other hand, from the optimization perspective, the two problems with or without shorting demonstrate very different features, as evidenced from our numerical experiments using CPLEX, and thus they are both worth studying. We also pay particular attention to the two special cases of CCMV in this paper, namely, the cardinality constrained Sharpe ratio maximization problem and the cardinality constrained global minimum-variance portfolio problem, because these two problems represent two important applications of the mean-variance portfolio selection model. The Sharpe ratio (see Sharpe 2007) measures the risk-adjusted performance, i.e., the excess returns per unit of risk. Due to its simple structure, the Sharpe ratio has become one of the standard performance measures in the financial industry. Recently, Jagannathan and Ma (2003) and DeMiguel et al. (2007) have shown that, due to the estimation error on the expected returns, the global minimum-variance portfolio is much more reliable when compared with general mean-variance portfolio policies, especially when the out-of-sample performance is examined.

The cardinality constrained mean-variance portfolio selection problem is a special case of general cardinality constrained quadratic optimization (CCQO) problems, which is known to be NP-hard. CCQO has been studied in the literature by several researchers. Bienstock (1996) considers such a problem with polyhedral constraints and upper bounds on decision variables. To deal with the cardinality constraint, Bienstock (1996) proposes to use a surrogate constraint to replace the cardinality constraint. Considering the same type of problems, Bertsimas and Shioda (2009) develop an exact solution approach by using a convex relaxation resulted from ignoring the cardinality constraint. Such a relaxation leads to a formulation ready for Lemke's pivoting method, which is a "simplex"-type method for solving quadratic programming problems. While integrating such a bounding method into a branch-and-bound algorithm, the "warm-start" strategy is adopted in each step, which accelerates the entire solution algorithm. Li et al. (2006) consider in their study several discrete features in the real trade practice, including the cardinality constraint. Utilizing the monotonicity of the zero norm corresponding to the cardinality constraint and other geometric characteristics of the quadratic objective contour, they develop some cutting schemes to reduce the feasible region successively in a branch-and-bound solution process. Chang et al. (2000) consider heuristic methods for the mean-variance portfolio selection problem with a cardinality constraint and integer constraints. Blog et al. (1983) propose a dynamic programming heuristic which is applicable only to small-scale portfolio selection problems. Shaw et al. (2008) use the Lagrangian relaxation to construct some lower bounds for such problems. Adopting a different solution concept,

Xie et al. (2008) adopt a randomized algorithm to identify good feasible solutions to CCMV problem with some quality guarantee. Recently, Gao and Li (2011) derive the optimal control law for discrete-time cardinality constrained linear-quadratic control problem, in which the number of time periods where controls can be applied is limited, by using a semi-definite programming solution scheme.

The study on CCQO in the literature has been carried out primarily for general cardinality constrained quadratic programming problem formulations with polyhedral constraints and upper and lower bounds on decision variables. The main solution scheme in the literature is to propose various approaches in relaxing the hard cardinality constraint. Motivated by the special structure of CCMV problems, we choose a different solution scheme. Applying the Lagrangian relaxation method to the primal problem, we can concentrate on a pure cardinality constrained quadratic optimization problem, which possesses a symmetric property and to which geometric approaches can be developed. In contrast to the existing literature that has focused primarily on some direct relaxations of the cardinality constraint, we consider modifying the objective function to some separable relaxations, which are immune to the hard cardinality constraint. Invoking certain discovered elegant problem features, we develop efficient lower bounding schemes by using the circumscribed box, the circumscribed ball, and the circumscribed axis-aligned ellipsoids to approximate the objective contour of the problem. In particular, all these cardinality constrained relaxation problems can be solved analytically. Furthermore, we derive efficient polynomial-time algorithms for the corresponding dual search problems. Most promisingly, the lower bounding scheme using the circumscribed axis-aligned ellipsoid leads to a semidefinite programming (SDP) formulation and offers a sharp bound and high-quality feasible solution. By integrating these lower bounding schemes into a branch-and-bound algorithm (BnB), our developed solution scheme outperforms CPLEX significantly in identifying the exact optimal portfolio. We would like to make one more point here. As far as we know, the literature lacks a study on pure CCQO (with only cardinality constraint), except a study by Gao and Li (2013) on a polynomially solvable case of pure CCQO. Our numerical tests show, however, that the existing relaxation methods developed for general CCQO problems with polyhedral constraints in the literature do not show their effectiveness for pure CCQO problems. Furthermore, both the cardinality constrained Sharpe ratio maximization problem and the cardinality constrained minimum-variance portfolio problem can be formulated equivalently as pure CCQO problems. Thus, our research also constitutes a meaningful work to provide efficient solution methods for pure CCQO problems.

To facilitate our presentation, we list in the following the problem formulations, which we investigate in this paper:

- ($\mathcal{P}_{\text{ccmv}}$): Cardinality constrained mean-variance portfolio selection problem (CCMV);

- $(\mathcal{P}_{\text{cnt}})$: The continuous relaxation of $(\mathcal{P}_{\text{ccmv}})$ by ignoring the cardinality constraint in $(\mathcal{P}_{\text{ccmv}})$;
- $(\mathcal{P}_{\text{mip}})$: Mixed-integer programming (MIP) formulation of $(\mathcal{P}_{\text{ccmv}})$;
- $(\mathcal{P}_{\text{mipcnt}})$: Continuous relaxation of $(\mathcal{P}_{\text{mip}})$;
- $(\mathcal{P}_{\text{sur}})$: The surrogate relaxation of $(\mathcal{P}_{\text{ccmv}})$ by replacing the cardinality constraint by a surrogate constraint;
- $(\mathcal{P}_{\text{ccqo}}^p)$: Pure cardinality constrained quadratic optimization problem;
- $(\mathcal{P}_{\text{box}})$, $(\mathcal{P}_{\text{ball}})$ and $(\mathcal{P}_{\text{ellp}})$: Relaxation of $(\mathcal{P}_{\text{ccqo}}^p)$ via replacing the objective contour of $(\mathcal{P}_{\text{ccqo}}^p)$, respectively, by circumscribed hyper-box, by circumscribed ball, and by circumscribed axis-aligned ellipsoid;
- $(\mathcal{P}_{\text{socp}})$: Second-order cone programming formulation resulted from $(\mathcal{P}_{\text{ellp}})$ when fixing a parametric matrix;
- $(\mathcal{P}_{\text{ccSharpe}})$: Cardinality constrained Sharpe ration maximization problem; and
- $(\mathcal{P}_{\text{cc-minvar}})$: Cardinality constrained global minimum variance problem.

Adding a superscript “ns” to any of the above formulations means the same problem with no shorting constraint.

Our paper is organized as follows. After the introduction in this section, we give the problem formulation of $(\mathcal{P}_{\text{ccmv}})$ in §2. We then develop our solution scheme for $(\mathcal{P}_{\text{ccmv}})$ in §3. More specifically, applying Lagrangian relaxation to $(\mathcal{P}_{\text{ccmv}})$ yields a problem in a form of $(\mathcal{P}_{\text{ccqo}}^p)$. Our solution algorithm is then developed by (i) constructing efficient lower bounding schemes for parametric $(\mathcal{P}_{\text{ccqo}}^p)$ for given Lagrangian multipliers and (ii) deriving efficient dual search schemes with respect to the Lagrangian multipliers. In §4, we prove that both $(\mathcal{P}_{\text{ccSharpe}})$ and $(\mathcal{P}_{\text{cc-minvar}})$ are equivalent to $(\mathcal{P}_{\text{ccqo}}^p)$. Thus, the solution algorithm developed for $(\mathcal{P}_{\text{ccmv}})$ can be further enhanced for $(\mathcal{P}_{\text{ccSharpe}})$ and $(\mathcal{P}_{\text{cc-minvar}})$, as there is no need for dual search. We discuss how to get a good feasible solution in §5. We test our algorithms for various types of problems numerically in §6, and we are able to reach a conclusion that our developed solution scheme outperforms CPLEX significantly in identifying the optimal portfolio. We conclude our paper in §7 with some suggestions for future study. Due to the page limit, we have placed some of the proofs for the lemmas and theorems in the e-companion to this paper, available as supplemental material at <http://dx.doi.org/10.1287/opre.2013.1170>. We have also reported in the e-companion a comparison of various lower bounds developed in this paper and also the existing ones in the literature.

Throughout this paper, we use $M \succeq 0$ ($M \succ 0$) to denote a positive semidefinite (positive definite) matrix M , $M^\dagger$ to denote the Moore-Penrose pseudoinverse of matrix M , $\text{diag}(a)$ to denote the diagonal matrix with its diagonal being vector a , $v(\cdot)$ to denote the optimal value of particular problem $(\cdot)$, I to denote the identity matrix and $\mathbf{1}$ to denote the vector with all elements being 1.

2. Problem Formulation

Suppose there are n risky assets in the asset pool, with their expected return vector given by $r \triangleq (r_1, \dots, r_n)'$ and their covariance matrix given by $D \triangleq \{D_{i,j}\}_{i,j=1}^n$, which is assumed to be strictly positive definite. Let $x = (x_1, \dots, x_n)'$ be the portfolio weights to the n risky assets such that $x \in \mathbb{R}^n$ and $\mathbf{1}'x = 1$. In real trading practice, however, almost no individual or financial institution intends to invest in all available assets. Due to the transaction cost, budget constraints, or even mental costs, investors, in general, confine themselves to selecting a *small* number of risky assets in forming a portfolio, i.e., selecting s assets from among total n assets to construct a portfolio. Mathematically, investors construct their portfolios from the following set of cardinality s with $s \in \mathbb{Z}_+$ and $s < n$,

$$\Delta(s) \triangleq \left\{ x \in \mathbb{R}^n \mid \sum_{i=1}^n \delta(x_i) \leq s \right\}, \quad (1)$$

where $\delta(\cdot)$ is the indicator function, i.e., $\delta(y) = 0$ if $y = 0$ and $\delta(y) = 1$ otherwise. The cardinality constrained mean-variance portfolio selection problem (CCMV) is then formulated as follows:

$$(\mathcal{P}_{\text{ccmv}}): \quad \min_{x \in \Delta(s)} \quad \frac{1}{2} x' D x, \quad (2)$$

$$\begin{aligned} \text{subject to} \quad & r'x = \bar{r}, \\ & \mathbf{1}'x = 1, \end{aligned} \quad (3)$$

where $\bar{r}$ is a target level of portfolio return. In the following discussion, we call vector $(\delta(x_1), \delta(x_2), \dots, \delta(x_n))' \in \{0, 1\}^n$ the *sparsity* of portfolio x . Note that once the sparsity of x is fixed, problem $(\mathcal{P}_{\text{ccmv}})$ can be solved explicitly.

Note that the proof of Theorem 1 in Bienstock (1996) for the NP-hardness of general CCQO problem is not applicable for problem $(\mathcal{P}_{\text{ccmv}})$. In particular, different from the general CCQO problem, finding a feasible solution for problem $(\mathcal{P}_{\text{ccmv}})$ is an easy problem. However, we still have the following complexity result.

LEMMA 1. *Problem $(\mathcal{P}_{\text{ccmv}})$ is NP-hard.*

PROOF. Please see the proof in the e-companion. $\square$

Ignoring the cardinality constraint in $(\mathcal{P}_{\text{ccmv}})$ gives rise to its continuous relaxation,

$$(\mathcal{P}_{\text{cnt}}): \quad \min_x \left\{ \frac{1}{2} x' D x : r'x = \bar{r}, \mathbf{1}'x = 1 \right\}.$$

As D is positive definite, the optimal portfolio of problem $(\mathcal{P}_{\text{cnt}})$ is $x_{\text{cnt}}^* = \alpha_1 D^{-1} r + \alpha_2 D^{-1} \mathbf{1}$, where $\begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = C^{-1} \begin{pmatrix} \bar{r} \\ 1 \end{pmatrix}$ and $C = \begin{pmatrix} r' D^{-1} r & r' D^{-1} \mathbf{1} \\ \mathbf{1}' D^{-1} r & \mathbf{1}' D^{-1} \mathbf{1} \end{pmatrix}$, with the optimal value $v(\mathcal{P}_{\text{cnt}}) = \frac{1}{2} \begin{pmatrix} \bar{r} \\ 1 \end{pmatrix}' C^{-1} \begin{pmatrix} \bar{r} \\ 1 \end{pmatrix}$. Although such a bound can be computed efficiently in $O(n^3)$ time (mainly for matrix inversion and multiplication), it could be quite loose for most situations.

We consider next the following equivalent formulation of problem $(\mathcal{P}_{\text{ccmv}})$, which takes a form of mixed-integer programming (MIP) (see, e.g., Bertsimas and Shioda 2009),

$$(\mathcal{P}_{\text{mip}}): \quad \min_x \quad \frac{1}{2}x'Dx, \\ \text{subject to} \quad \mathbf{1}'x = 1, \quad r'x = \bar{r}, \\ -Mz_i \leq x_i \leq z_i M, \quad i = 1, \dots, n, \\ \sum_{i=1}^n z_i \leq s, \quad z_i \in \{0, 1\}, \quad i = 1, \dots, n,$$

where $M > 0$ is a large number that represents an upper bound for the absolute value of all elements in the optimal solution to problem $(\mathcal{P}_{\text{mip}})$. Such a formulation $(\mathcal{P}_{\text{mip}})$ can be directly solved by any mixed-integer solver such as CPLEX solver (CPLEX.12.3 2011). The formulation $(\mathcal{P}_{\text{mip}})$, however, does suffer from some undesired properties, which we state below. If we use a branch-and-bound algorithm as a solution scheme for problem $(\mathcal{P}_{\text{mip}})$, one natural relaxation is to relax $z_i \in \{0, 1\}$ to $z_i \in [0, 1]$, $i = 1, \dots, n$, resulting in another version of *continuous relaxation* denoted by $(\mathcal{P}_{\text{mipent}})$ in our following discussion. The bound quality of $(\mathcal{P}_{\text{mipent}})$ depends heavily on parameter M . On one hand, M should be chosen large enough such that the real optimal solution is contained in the box $[-M, M]^n$. On the other hand, M cannot be too large because it leads to a very loose lower bound generated by $(\mathcal{P}_{\text{mipent}})$.

LEMMA 2. *If $y^* = (y_1^*, \dots, y_n^*)'$ solves problem $(\mathcal{P}_{\text{cnt}})$ and M is chosen such that $M \geq (1/s) \sum_{i=1}^n |y_i^*|$ and $M \geq \max_{i=1, \dots, n} |y_i^*|$, then $v(\mathcal{P}_{\text{mipent}}) = v(\mathcal{P}_{\text{cnt}})$.*

PROOF. Please see the proof in the e-companion. $\square$

Lemma 2 suggests that the continuous relaxation generated from the mixed-integer formulation, $(\mathcal{P}_{\text{mipent}})$, does not ensure a better bound when compared with the trivial continuous relaxation by simply ignoring cardinality constraint (1) in problem $(\mathcal{P}_{\text{ccmv}})$. From the geometric point of view, if we let $z_i = |y_i^*|/M$, for $i = 1, \dots, n$, we have $\sum_{i=1}^n |x_i|/M \leq s$ under the conditions in Lemma 2. Note that $\sum_{i=1}^n |y_i^*|/M \leq s$ is the surrogate constraint proposed in Bienstock (1996), which is a polyhedral constraint and serves to relax the original cardinality constraint. Under the conditions in Lemma 2, solution y^* always satisfies this polyhedral constraint, thus making the relaxation $(\mathcal{P}_{\text{mipent}})$ perform similar to trivial continuous relaxation $(\mathcal{P}_{\text{cnt}})$. Thus, Lemma 2 also indicates certain conditions under which the surrogate constraint method (Bienstock 1996) does not perform better than the continuous relaxation. Although Lemma 2 is a sufficient condition for $v(\mathcal{P}_{\text{cnt}}) = v(\mathcal{P}_{\text{mipent}})$, we find in our numerical tests (see §6) that the bounds from $(\mathcal{P}_{\text{mipent}})$ and $(\mathcal{P}_{\text{cnt}})$ are almost identical for all our testing problems.

In this research, we will develop our solution scheme centering on the following pure cardinality constrained

quadratic optimization problem $(\mathcal{P}_{\text{ccqo}}^p)$,

$$(\mathcal{P}_{\text{ccqo}}^p(D, \bar{d}, \bar{c})): \\ \min_x \{f(x) \triangleq \frac{1}{2}x'Dx + \bar{d}'x + \bar{c} : x \in \Delta(s)\}, \quad (4)$$

where D is assumed, without loss of generality, to be the same as the one in $(\mathcal{P}_{\text{ccmv}})$ and $\bar{d} \in \mathbb{R}^n$ and $\bar{c} \in \mathbb{R}$ are a given vector and a given scalar, respectively. Note that applying the Lagrangian relaxation to problem $(\mathcal{P}_{\text{ccmv}})$ results in a pure cardinality constrained portfolio selection problem in the form of $(\mathcal{P}_{\text{ccqo}}^p)$. Furthermore, we will show in §4 that the two important portfolio selection problems, the Sharpe ratio maximization problem and the global variance minimization problem, are equivalent to $(\mathcal{P}_{\text{ccqo}}^p)$.

3. Solution of Problem $(\mathcal{P}_{\text{ccmv}})$

3.1. Geometric Approaches

Now, we examine problem $(\mathcal{P}_{\text{ccmv}})$ from another angle. Introducing the Lagrangian multipliers $\eta \in \mathbb{R}$ and $\mu \in \mathbb{R}$ for constraints (2) and (3), respectively, in problem $(\mathcal{P}_{\text{ccmv}})$ yields the following Lagrangian function:

$$L(x, \eta, \mu) \triangleq \frac{1}{2}x'Dx + \eta(\bar{r} - r'x) + \mu(\mathbf{1}'x - 1) \\ = \frac{1}{2}x'Dx + d(\eta, \mu)'x + c(\eta, \mu), \quad (5)$$

where

$$d(\eta, \mu) \triangleq \mu \mathbf{1} - \eta r, \quad c(\eta, \mu) \triangleq \eta \bar{r} - \mu. \quad (6)$$

For any $\eta \in \mathbb{R}$ and $\mu \in \mathbb{R}$, we consider now the following Lagrangian relaxation, which is exactly a special case of $(\mathcal{P}_{\text{ccqo}}^p)$ defined in §2:

$$(\mathcal{P}_{\text{ccqo}}^p(D, d(\eta, \mu), c(\eta, \mu))) : \\ \min_x \quad L(x, \eta, \mu) = \frac{1}{2}x'Dx + d'(\eta, \mu)x + c(\eta, \mu), \\ \text{subject to} \quad \sum_{i=0}^n \delta(x_i) \leq s.$$

Clearly, it holds that $v(\mathcal{P}_{\text{ccqo}}^p(D, d(\eta, \mu), c(\eta, \mu))) \leq v(\mathcal{P}_{\text{ccmv}})$ for any $\eta \in \mathbb{R}$ and $\mu \in \mathbb{R}$. Our main solution scheme is to first construct the lower bounds of problem $(\mathcal{P}_{\text{ccqo}}^p(D, d(\eta, \mu), c(\eta, \mu)))$, and then to strengthen such lower bounds by using the dual search methods with respect to η and μ in order to generate tight lower bounds for problem $(\mathcal{P}_{\text{ccmv}})$. We denote $d(\eta, \mu)$ and $c(\eta, \mu)$ by d and c , respectively, in the following discussion.

Since $D \succ 0$, geometrically speaking, the objective contour of problem $(\mathcal{P}_{\text{ccqo}}^p(D, d, c))$ is an ellipsoid in $\mathbb{R}^n$ space. We term such a property in a pure cardinality constrained quadratic optimization formulation as a symmetric property, which in turn offers an advantage for us to adopt a

geometric solution approach. For any $P \succeq 0$ and $p \in \mathbb{R}^n$, we denote the following ellipsoid in $\mathbb{R}^n$ as

$$\mathcal{E}(P, p, \rho) \triangleq \{y \in \mathbb{R}^n \mid (y + p)' P (y + p) \leq \rho\}, \quad (7)$$

where $\rho \geq 0$. Thus, solving problem $(\mathcal{P}_{\text{ccqo}}^p(D, d, c))$ is equivalent to finding the minimum ellipsoid that touches the set $\Delta(s)$, i.e.,

$$(\mathcal{P}_{\text{ccqo}}^p(D, d, c)) : \min_{x \in \Delta(s), \tau} \tau, \quad \text{subject to } x \in \mathcal{E}(D, l, \rho(\tau)), \quad (8)$$

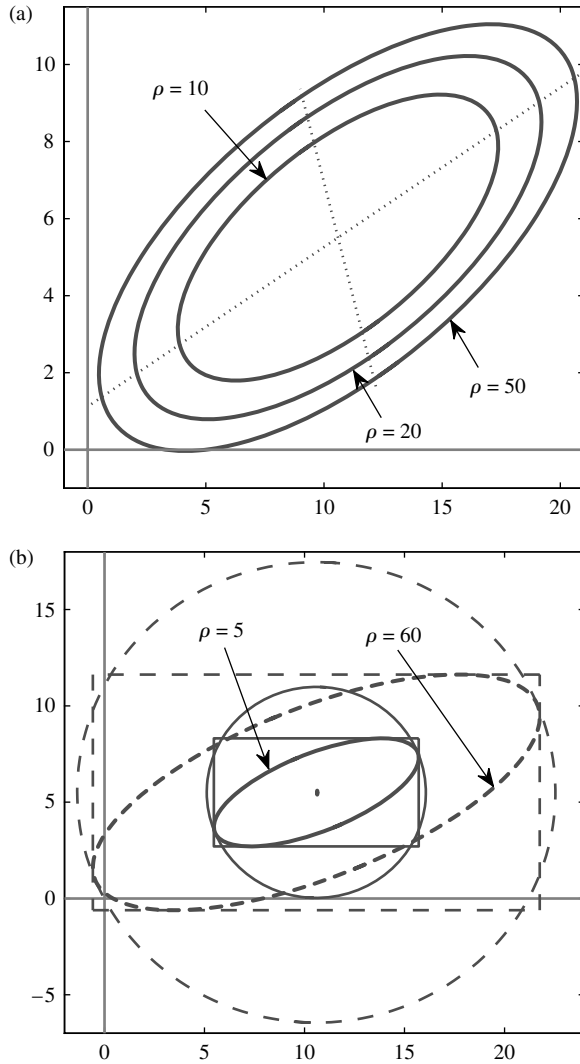
where $\mathcal{E}(D, l, \rho(\tau)) \triangleq \{x \in \mathbb{R}^n \mid L(x, \eta, \mu) \leq \tau\} = \{x \in \mathbb{R}^n \mid (x + l)' D (x + l) \leq \rho(\tau)\}$ with

$$l = (l_1, \dots, l_n)' \triangleq D^{-1}d = \mu D^{-1}\mathbf{1} - \eta D^{-1}r, \quad (9)$$

$$\rho(\tau) \triangleq 2\tau + d' D^{-1}d - 2c. \quad (10)$$

An illustrative example of problem $(\mathcal{P}_{\text{ccqo}}^p(D, d, c))$ with $n = 2$ and $s = 1$ is shown in Figure 1(a). The optimal value of problem $(\mathcal{P}_{\text{ccqo}}^p(D, d, c))$ is achieved when the contour touches the x -axis.

Figure 1. The objective contour and its relaxations.



3.2. Lower Bounds Generated from Circumscribed Hyper-Box and Circumscribed Ball

The symmetric property exhibited in $(\mathcal{P}_{\text{ccqo}}^p(D, d, c))$ enables us to explore the geometric features behind the mathematical formulation. We derive first lower bounds of problem $(\mathcal{P}_{\text{ccqo}}^p(D, d, c))$ in this subsection using the circumscribed hyper-box and circumscribed ball. Denote the hyper-box in $\mathbb{R}^n$ by

$$[\alpha^l, \alpha^u] \triangleq \{(y_1, \dots, y_n)' \mid \alpha_i^l \leq y_i \leq \alpha_i^u, i = 1, \dots, n\},$$

where $\alpha^l = (\alpha_1^l, \dots, \alpha_n^l)'$ and $\alpha^u = (\alpha_1^u, \dots, \alpha_n^u)'$ are the lower and upper corner points of the hyper-box, respectively. The following result is true (see Li and Sun 2006).

LEMMA 3. *The circumscribed hyper-box of ellipsoid $\mathcal{E}(D, l, \rho(\tau))$ is given by $[\alpha^l(\tau), \alpha^u(\tau)] := \{y \in \mathbb{R}^n \mid \alpha^l(\tau) \leq y \leq \alpha^u(\tau)\}$, where*

$$\alpha^l(\tau) = -l - \sqrt{\rho(\tau)} \text{vec}\{\sqrt{\hat{D}_i}\},$$

$$\alpha^u(\tau) = -l + \sqrt{\rho(\tau)} \text{vec}\{\sqrt{\hat{D}_i}\},$$

and $\text{vec}\{\sqrt{\hat{D}_i}\} := (\sqrt{\hat{D}_1} \sqrt{\hat{D}_2} \dots \sqrt{\hat{D}_n})'$ with $\hat{D}_i$ being the i th component of the diagonal of D^{-1} .

The above explicit expression of such a circumscribed hyper-box motivates us to relax the objective contour of problem $(\mathcal{P}_{\text{ccqo}}^p(D, d, c))$. In particular, we relax the ellipsoid constraint (8) to its minimum circumscribed hyper-box $[\alpha^l(\tau), \alpha^u(\tau)]$ in the following form:

$$(\mathcal{P}_{\text{box}}(\eta, \mu)) : \min_{x \in \Delta(s), \tau} \tau, \quad \text{subject to } \frac{(x_i + l_i)^2}{\hat{D}_i} \leq \rho(\tau), \quad i = 1, \dots, n. \quad (11)$$

Since $\mathcal{E}(D, l, \rho(\tau)) \subset [\alpha^l(\tau), \alpha^u(\tau)]$ for any τ , we have $v(\mathcal{P}_{\text{box}}(\eta, \mu)) \leq v(\mathcal{P}_{\text{ccqo}}^p(D, d, c))$. More importantly, problem $(\mathcal{P}_{\text{box}}(\eta, \mu))$ can be solved explicitly.

THEOREM 1. *For any fixed η and μ , the optimal value of problem $(\mathcal{P}_{\text{box}}(\eta, \mu))$ is*

$$v(\mathcal{P}_{\text{box}}(\eta, \mu)) = \frac{(l_{i^*})^2}{2\hat{D}_{i^*}} - \frac{1}{2}d' D^{-1}d + c, \quad (12)$$

where i^* is the index of the $(s + 1)$ th largest elements of $\{(l_i)^2 / \hat{D}_i\}_{i=1}^n$.

PROOF. Suppose that $x_i = 0$ for some i , the constraint indexed by i in (11) gives rise to $(l_i)^2 / \hat{D}_i \leq \rho(\tau) = 2\tau + d' D^{-1}d - 2c$. Thus, if all x_i s take value 0, the minimum τ corresponding to this set of inequalities is the maximum among $(l_i)^2 / (2\hat{D}_i) - \frac{1}{2}d' D^{-1}d + c$, for $i = 1, \dots, n$.

As the cardinality constraint (1) allows us to set at most s elements of x_i to be nonzero, we can set x_i equal to $-l_i$ for those i corresponding to the s largest elements of $(l_i)^2/(2\hat{D}_i) - \frac{1}{2}d'D^{-1}d + c$. Thus, the minimum of τ becomes the $(s+1)$ th largest element of $(l_i)^2/(2\hat{D}_i) - \frac{1}{2}d'D^{-1}d + c$ among all $i = 1, \dots, n$. $\square$

Using a similar idea, we can relax the constraint (8) to its minimum circumscribed ball, giving rise to the following relaxation:

$$(\mathcal{P}_{\text{ball}}(\eta, \mu)): \min_{x \in \Delta(s), \tau} \tau, \quad \text{subject to } \lambda_{\min}^D \sum_{i=1}^n (x_i + l_i)^2 \leq \rho(\tau), \quad (13)$$

where $\lambda_{\min}^D$ is the minimum eigenvalue of the matrix D . Because $\mathcal{E}(D, l, \rho(\tau)) \subset \mathcal{E}(\lambda_{\min}^D I, l, \rho(\tau))$, we always have $v(\mathcal{P}_{\text{ball}}(\eta, \mu)) \leq v(\mathcal{P}_{\text{ccmv}})$.

THEOREM 2. *The optimal value of problem $(\mathcal{P}_{\text{ball}}(\eta, \mu))$ is*

$$v(\mathcal{P}_{\text{ball}}(\eta, \mu)) = \frac{1}{2} \lambda_{\min}^D \sum_{i \in \mathcal{J}_{n-s}} (l_i)^2 - \frac{1}{2} d' D^{-1} d + c, \quad (14)$$

where $\mathcal{J}_{n-s}$ denotes the first $n - s$ smallest elements of $\{l_i^2\}_{i=1}^n$.

PROOF. Since $x \in \Delta(s)$, we can set at most s nonzero elements of x to cancel $\{l_i\}_{i=1}^n$ in order to minimize $\rho(\tau)$. It becomes clear that the best choice is to choose x to cancel the s largest elements of $\{l_i^2\}_{i=1}^n$. $\square$

For some fixed η and μ , we demonstrate the bounds generated from a circumscribed box or a circumscribed ball in an illustrative example plotted in Figure 1(b).

Following Theorems 1 and 2, the lower bounds from $(\mathcal{P}_{\text{box}}(\eta, \mu))$ and $(\mathcal{P}_{\text{ball}}(\eta, \mu))$ are derived, respectively, for fixed η and μ . These lower bounds can be further strengthened by performing a dual search procedure:

$$(\mathcal{P}_{\text{box}}): \max_{\eta, \mu} v(\mathcal{P}_{\text{box}}(\eta, \mu)) = \frac{(l_{i^*})^2}{2\hat{D}_{i^*}} - \frac{1}{2} d' D^{-1} d + c, \quad (15)$$

$$(\mathcal{P}_{\text{ball}}): \max_{\eta, \mu} v(\mathcal{P}_{\text{ball}}(\eta, \mu)) = \frac{1}{2} \lambda_{\min}^D \sum_{i \in \mathcal{J}_{n-s}} (l_i^2) - \frac{1}{2} d' D^{-1} d + c, \quad (16)$$

where i^* is the index of the $(s+1)$ th largest elements of $\{(l_i)^2/\hat{D}_i\}_{i=1}^n$ and $\mathcal{J}_{n-s}$ denotes the first $n - s$ smallest elements of $\{l_i^2\}_{i=1}^n$. The detailed dual search procedures for solving problems $(\mathcal{P}_{\text{box}})$ and $(\mathcal{P}_{\text{ball}})$ will be discussed in §3.3. One significant finding is that the bounds derived from $(\mathcal{P}_{\text{box}})$ and $(\mathcal{P}_{\text{ball}})$ are both strictly better than the continuous relaxation.

THEOREM 3. *If $D > 0$ in problem $(\mathcal{P}_{\text{ccmv}})$, then $v(\mathcal{P}_{\text{ccmv}}) \geq v(\mathcal{P}_{\text{box}}) > v(\mathcal{P}_{\text{cnt}})$ and $v(\mathcal{P}_{\text{ccmv}}) \geq v(\mathcal{P}_{\text{ball}}) > v(\mathcal{P}_{\text{cnt}})$.*

PROOF. As problem $(\mathcal{P}_{\text{cnt}})$ is a convex optimization problem, the strong duality relationship holds, i.e.,

$$v(\mathcal{P}_{\text{cnt}}) = \max_{\eta, \mu} \min_x L(x, \mu, \eta) = \max_{\eta, \mu} -\frac{1}{2} d' D^{-1} d + c, \quad (17)$$

where $L(x, \eta, \mu)$ is defined in (5). For any fixed $\eta \in \mathbb{R}$ and $\mu \in \mathbb{R}$, comparing (17) with (12) and (14) respectively yields

$$v(\mathcal{P}_{\text{box}}(\eta, \mu)) > -\frac{1}{2} d' D^{-1} d + c, \quad \forall \eta \in \mathbb{R}, \mu \in \mathbb{R}, \quad (18)$$

$$v(\mathcal{P}_{\text{ball}}(\eta, \mu)) > -\frac{1}{2} d' D^{-1} d + c, \quad \forall \eta \in \mathbb{R}, \mu \in \mathbb{R}. \quad (19)$$

Taking maximum with respect to η and μ on both sides of (18) and (19) yields the conclusion of the theorem. $\square$

3.3. The Dual Search for Problems

$(\mathcal{P}_{\text{box}})$ and $(\mathcal{P}_{\text{ball}})$

In this section, we propose a solution method to solve both the dual problems, (15) and (16). In both problems, we have to identify the order of a class of functions, i.e., the order of functions $\{l_i^2/\hat{D}_i\}_{i=1}^n$ and $\{l_i^2\}_{i=1}^n$ for problems (15) and (16), respectively. To simplify the illustration, we explain and solve problem (15) first. Note that to solve box-bound (15), we need to identify the $(n+1)$ th largest elements of $\{l_i^2/\hat{D}_i\}_{i=1}^n$. However, the order of the functions $\{l_i^2/\hat{D}_i\}_{i=1}^n$ varies while the value of $\eta \in \mathbb{R}$ and $\mu \in \mathbb{R}$ vary. We introduce the following notations.

$$a = (a_1, \dots, a_n)' \triangleq D^{-1} r,$$

$$u = (u_1, \dots, u_n)' \triangleq D^{-1} \mathbf{1},$$

$$\hat{a}_i \triangleq a_i / \sqrt{\hat{D}_i},$$

$$\hat{u}_i \triangleq u_i / \sqrt{\hat{D}_i},$$

for $i = 1, \dots, n$. Since when $\eta = 0$, the order of functions $\{l_i^2/\hat{D}_i\}_{i=1}^n$ is determined by $|u_i|$, we need only to consider the case of $\eta \neq 0$. The definition (9) implies that comparing two items $l_i^2/\hat{D}_i$ and $l_j^2/\hat{D}_j$ for $i \neq j$ is equivalent to checking whether the difference

$$l_i^2/\hat{D}_i - l_j^2/\hat{D}_j = \eta^2 [(\hat{a}_i - \hat{u}_i \mu / \eta)^2 - (\hat{a}_j - \hat{u}_j \mu / \eta)^2] \quad (20)$$

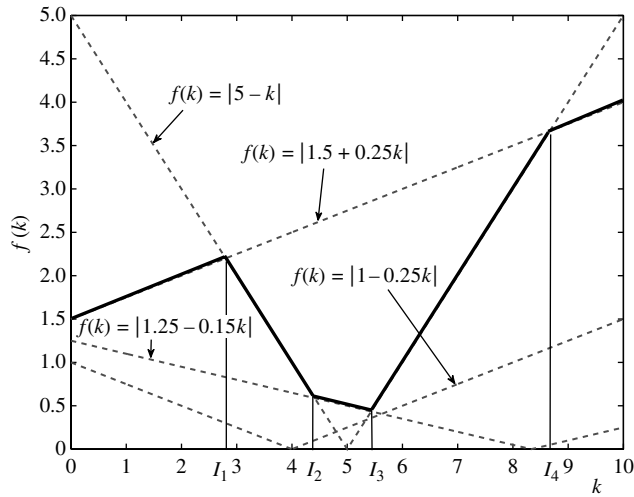
is greater or less than 0. Thus, for any η and μ , the order of functions $\{l_i^2/\hat{D}_i\}_{i=1}^n$ is the same as that of functions $\{f_i(\kappa)\}_{i=1}^n$, where

$$f_i(\kappa) \triangleq |\hat{a}_i - \kappa \hat{u}_i|, \quad i = 1, \dots, n, \quad (21)$$

with $\kappa = \mu/\eta$. Clearly, $f_i(\kappa)$ is a V-shape piecewise linear function. Next, we consider the following breaking points $\kappa_{i,j}^1, \kappa_{i,j}^2, i, j \in \mathbb{I} = \{i = 1, \dots, n-1, j = i+1, \dots, n\}$,

$$f_i(\kappa) - f_j(\kappa) = 0 \rightarrow \begin{cases} \kappa_{i,j}^1 = (\hat{a}_i + \hat{a}_j)/(\hat{u}_i + \hat{u}_j) & \text{if } \hat{u}_i + \hat{u}_j \neq 0, \\ \kappa_{i,j}^2 = (\hat{a}_i - \hat{a}_j)/(\hat{u}_i - \hat{u}_j) & \text{if } \hat{u}_i - \hat{u}_j \neq 0. \end{cases} \quad (22)$$

Figure 2. The second-largest function of $f_i(\kappa)$.



Obviously, as κ varies in $[-\infty, \infty]$, the necessary condition that the order of functions $\{f_i(\kappa)\}_{i=1}^n$ changes is that κ hits the breaking points $\{\kappa_{i,j}^1, \kappa_{i,j}^2\}_{i,j \in \mathbb{I}}$. Suppose there are total N breaking points. Arranging these points $\{\kappa_{i,j}^1, \kappa_{i,j}^2\}_{i,j \in \mathbb{I}}$ in an ascending order partitions the real line to $N + 1$ intervals $[I_0, I_1], [I_1, I_2], \dots, [I_{N-1}, I_N], [I_N, I_{N+1}]$, where $I_0 = -\infty$ and $I_{N+1} = \infty$.

Let us consider an example with $n = 4$, $f_1(\kappa) = |1.5 + 0.25\kappa|$, $f_2(\kappa) = |5 - \kappa|$, $f_3(\kappa) = |1.25 - 0.15\kappa|$ and $f_4(\kappa) = |1 - 0.25\kappa|$. If we want to identify the second-largest functions of $\{f_i(\kappa)\}_{i=1}^4$, we have only to identify the breaking points that are plotted in Figure 2. In Figure 2, the bold line represents the second-largest function among $\{f_i(\kappa)\}_{i=1}^4$.

LEMMA 4. The total number, N , of breaking points defined in (22) in interval $[-\infty, \infty]$ is upper-bounded by $n(n - 1)$.

PROOF. All the breaking points in $[-\infty, \infty]$ can be computed. Let S_k be the number of breaking points for k functions of f_i . When $k = 2$, it holds that $S_2 \leq 2$, and, when $k > 3$, the recursion $S_k \leq S_{k-1} + 2(k - 1)$ holds. Solving such a recursion yields $S_n \leq (n - 1)n$. Thus, the total number of breaking points in $[-\infty, \infty]$, N , satisfies $N = S_n \leq n(n - 1)$. $\square$

Note that in each interval $[t_i, t_{i+1}]$ of κ , the orders of functions $\{f_i(\kappa)\}_{i=1}^n$ and $\{l_i^2/\hat{D}_j\}$ are invariant. Note further that we are interested only in finding those breaking points where i^* , the index of the $(s + 1)$ th largest element of $\{f_i(\kappa)\}_{i=1}^n$, changes. Refer to Figure 2, we are interested only in these breaking points on the bold line. Recognizing this fact helps us reduce further the computational efforts, although we have already proved that the total number of the breaking points is polynomially bounded by $n(n - 1)$.

The dual problem (15) can now be solved by comparing the objective of at most N subproblems, with the following form for each interval $[I_t, I_{t+1}]$, $t = 0, \dots, N$:

$$(J_t) \quad \max_{\eta, \mu} \left\{ \frac{1}{2} \begin{pmatrix} \eta \\ \mu \end{pmatrix}' \bar{P}_t \begin{pmatrix} \eta \\ \mu \end{pmatrix} + \begin{pmatrix} \bar{r} \\ -1 \end{pmatrix} \begin{pmatrix} \eta \\ \mu \end{pmatrix} : I_t \leq \mu/\eta \leq I_{t+1} \right\}, \quad (23)$$

where $\bar{P}_t$ is $\{f_i(\kappa)\}$ order-dependent and is given by

$$\bar{P}_t = \frac{1}{\hat{D}_{i^*}} \begin{pmatrix} a_{i^*}^2 & -a_{i^*} u_{i^*} \\ -a_{i^*} u_{i^*} & u_{i^*}^2 \end{pmatrix} - \begin{pmatrix} r'D^{-1}r & -r'D^{-1}\mathbf{1} \\ -\mathbf{1}'D^{-1}r & \mathbf{1}'D^{-1}\mathbf{1} \end{pmatrix}, \quad (24)$$

with i^* being the index of the $(s + 1)$ th largest element of $\{f_i(\kappa)\}_{i=1}^n$. Note that (24) is derived from (15) and (9). In problem (23), the feasible region is the union of two convex cones, $\{\eta I_t \leq \mu \leq \eta I_{t+1}, \eta \geq 0\}$ and $\{\eta I_t \geq \mu \geq \eta I_{t+1}, \eta \leq 0\}$. Thus, we can solve these subproblems separately. The dual bound (15) is $v(\mathcal{P}_{\text{box}}) = \max_{t=0, \dots, N} v(J_t)$.

To compute the dual bound (16), we have to identify the first $n - s$ smallest elements of $\{l_i^2\}_{i=1}^n$. We have only to replace $\hat{a}_i$ and $\hat{u}_i$ in our above analysis by a_i and u_i , respectively, for $i = 1, \dots, n$, and compute the breaking points according to (22). Ranging the breaking points gives rise to the interval $[I_t, I_{t+1}]$ for $t = 0, \dots, N$. Similarly, we have the quadratic optimization problem in interval $[I_t, I_{t+1}]$ the same as (23),

$$(\hat{J}_t) \quad \max_{\eta, \mu} \left\{ \frac{1}{2} \begin{pmatrix} \eta \\ \mu \end{pmatrix}' \hat{P}_t \begin{pmatrix} \eta \\ \mu \end{pmatrix} + \begin{pmatrix} \bar{r} \\ -1 \end{pmatrix} \begin{pmatrix} \eta \\ \mu \end{pmatrix} : I_t \leq \mu/\eta \leq I_{t+1} \right\}, \quad (25)$$

where $\hat{P}_t$ is $\{l_i^2\}$ order-dependent and is given by

$$\hat{P}_t = \lambda_{\min}^D \sum_{i \in \mathcal{I}_{n-s}} \begin{pmatrix} a_i^2 & -a_i u_i \\ -a_i u_i & u_i^2 \end{pmatrix} - \begin{pmatrix} r'D^{-1}r & -r'D^{-1}\mathbf{1} \\ -\mathbf{1}'D^{-1}r & \mathbf{1}'D^{-1}\mathbf{1} \end{pmatrix}, \quad (26)$$

with $\mathcal{I}_{n-s}$ being the index set of the first $n - s$ elements of $\{l_i^2\}_{i=1}^n$. Note that (26) is derived from (16) and (9). Then the dual bound in (16) can be computed as $v(\mathcal{P}_{\text{ball}}) = \max_{t=0, \dots, N} v(\hat{J}_t)$.

THEOREM 4. Matrix $\hat{P}_t$ defined in (26) is negative definite.

PROOF. Without loss of generality, we assume that the first $n - s$ smallest elements of $\{l_i^2\}_{i=1}^n$ are the first $n - s$ indices. Let

$$D_{n-s} = \text{diag}(\underbrace{\lambda_{\min}^D, \dots, \lambda_{\min}^D}_{n-s}, 0, \dots, 0).$$

From (16), the quadratic term can be rewritten as

$$\begin{aligned} \begin{pmatrix} \eta \\ \mu \end{pmatrix}' \hat{P}_t \begin{pmatrix} \eta \\ \mu \end{pmatrix} &= \lambda_{\min}^D \sum_{i=1}^{n-s} l_i^2 - d'D^{-1}d \\ &= l'D_{n-s}l - l'Dl = l'(D_{n-s} - D)l. \end{aligned}$$

Since $D_{n-s} - D \leq 0$ and l is linear with η and μ , we have $\hat{P}_l \leq 0$. $\square$

By using Theorem 4, we know that subproblem (25) is a two-dimensional concave maximization problem, which can be solved explicitly. However, $\bar{P}_l$ in subproblem (23) is not negative definite in general. We thus need to solve problem (23) by using some heuristics in order to achieve a good feasible solution. Nevertheless, because the feasible region of problem (23) is a two-dimensional cone, we can identify the optimal solution by checking an interior solution and the solutions on the two boundaries.

3.4. Lower Bounds Generated from Circumscribed Axis-Aligned Ellipsoid

Since the objective contour of problem $(\mathcal{P}_{\text{ccqo}}^p(D, d, c))$ is an ellipsoid, it is also natural to consider adopting some specially structured ellipsoids as an approximation to the objective contour. In particular, we approximate problem $(\mathcal{P}_{\text{ccqo}}^p(D, d, c))$ by adopting the following class of problem formulations:

$$(\mathcal{P}_e(H, h, \xi)): \min_x \{f_e(x) \triangleq \frac{1}{2}x'Hx + h'x + \xi: x \in \Delta(s)\}, \quad (27)$$

where matrix H is diagonal, i.e., $H \triangleq \text{diag}\{H_1, H_2, \dots, H_n\}$, with $H_i > 0$, $i = 1, \dots, n$, $h \triangleq (h_1, \dots, h_n)'$ and $\xi \in \mathbb{R}$. Clearly, for given H , h and ξ , the objective function $f_e(x)$ of problem $(\mathcal{P}_e(H, h, \xi))$ is separable with respect to each x_i and the objective contour of problem $(\mathcal{P}_e(H, h, \xi))$ is thus an axis-aligned ellipsoid. Such a separability leads to the explicit solution of problem $(\mathcal{P}_e(H, h, \xi))$.

LEMMA 5. Let $\mathcal{J}_s$ be the index set of the first s smallest elements of $\{-h_i^2/H_i\}_{i=1}^n$, then the optimal solution of problem $(\mathcal{P}_e(H, h, \xi))$ is

$$x_i^* = \begin{cases} -\frac{h_i}{H_i}, & \text{if } i \in \mathcal{J}_s, \\ 0, & \text{if } i \notin \mathcal{J}_s, \end{cases} \quad \text{for } i = 1, \dots, n,$$

with the optimal value being

$$v(\mathcal{P}_e(H, h, \xi)) = -\frac{1}{2} \sum_{i \in \mathcal{J}_s} \frac{h_i^2}{H_i} + \xi.$$

PROOF. Since $f_e(x) = \sum_{i=1}^n (\frac{1}{2}H_i x_i^2 + h_i x_i) + \xi$, minimizing $f_e(x)$ is equivalent to minimizing each $g_i(x_i) \triangleq \frac{1}{2}H_i x_i^2 + h_i x_i$, for $i = 1, \dots, n$. Clearly, the optimal value of $g_i(x_i)$ is $g_i(x_i^*) = -h_i^2/(2H_i) \leq 0$, where $x_i^* = -h_i/H_i$, $i = 1, \dots, n$. As x must be confined in $\Delta(s)$, the optimal value is achieved by selecting the first s smallest elements of $g_i(x_i^*)$. $\square$

The explicit solution of problem $(\mathcal{P}_e(H, h, \xi))$ motivates us to construct a lower bound of problem $(\mathcal{P}_{\text{ccqo}}^p(D, d, c))$.

If we treat all H , h , and ξ as the decision variables, the condition

$$\begin{pmatrix} D - H & d - h \\ (d - h)' & 2c - 2\xi \end{pmatrix} = \begin{pmatrix} D - H & \mu \mathbf{1} - \eta r - h \\ (\mu \mathbf{1} - \eta r - h)' & 2(\eta \bar{r} - \mu - \xi) \end{pmatrix} \succeq 0 \quad (28)$$

guarantees $L(x, \eta, \mu) - f_e(x) \geq 0$ for all $x \in \mathbb{R}^n$, where $L(x, \eta, \mu)$ is the objective function of problem $(\mathcal{P}_{\text{ccqo}}^p(D, d, c))$ defined in (5). Minimizing the gap between $v(\mathcal{P}_e(H, h, \xi))$ and $v(\mathcal{P}_{\text{ccqo}}^p(D, d, c))$ is equivalent to finding the optimal H , h and ξ of the following problem:

$$(\mathcal{P}_{\text{ellp}}(\eta, \mu)): \max_{H > 0, h, \xi} \min_x f_e(x) = \frac{1}{2}x'Hx + h'x + \xi, \quad \text{subject to } x \in \Delta(s), \quad \{H, h, \xi\} \text{ satisfies (28)}. \quad (29)$$

Problem $(\mathcal{P}_{\text{ellp}}(\eta, \mu))$ can be further reformulated as a semidefinite programming problem, which can be solved efficiently by interior point method (see Vandenberghe and Boyd 1996). Note that the following result is true.

LEMMA 6. Given a set of functions $g_i(x): \mathbb{R}^n \rightarrow \mathbb{R}$, for $i = 1, \dots, n$. Let $\mathcal{J}_s$ be the index set of the first s smallest functions of $\{g_i(x)\}_{i=1}^n$. Then, finding the maximum value of $\sum_{i \in \mathcal{J}_s} g_i(x)$ is equivalent to the following problem:

$$(\mathcal{P}_{\text{aux}}): \max_{x, z, b_i} \quad sz + \sum_{i=1}^n b_i, \quad \text{subject to } b_i \leq 0, \quad i = 1, \dots, n, \quad b_i \leq g_i(x) - z, \quad i = 1, \dots, n,$$

where b_i , $i = 1, \dots, n$, and z are auxiliary variables.

PROOF. We prove that $v(\mathcal{P}_{\text{aux}}) = \max_x \sum_{i \in \mathcal{J}_s} g_i(x)$. First, for any x , we can construct a feasible solution, $\{\hat{b}_i\}_{i=1}^n$ and $\hat{z}$, of problem $(\mathcal{P}_{\text{aux}})$ as follows: (i) $\hat{b}_i = g_i(x) - \hat{z}$, for $i \in \mathcal{J}_s$ and $\hat{b}_i = 0$ for $i \notin \mathcal{J}_s$, and (ii) $\hat{z}$ is equal to the s th smallest element of $g_i(x)$. Such a feasible solution gives rise to the objective value $s\hat{z} + \sum_{i \in \mathcal{J}_s} (g_i(x) - \hat{z})$. Then, for any x , we have

$$\sum_{i \in \mathcal{J}_s} g_i(x) = s\hat{z} + \sum_{i \in \mathcal{J}_s} (g_i(x) - \hat{z}) \leq v(\mathcal{P}_{\text{aux}}).$$

Thus, we have $\max_x \sum_{i \in \mathcal{J}_s} g_i(x) \leq v(\mathcal{P}_{\text{aux}})$. On the other hand, in problem $(\mathcal{P}_{\text{aux}})$, since $b_i \leq 0$ and $b_i \leq g_i(x) - z$, we have

$$\begin{aligned} sz + \sum_{i=1}^n b_i &\leq sz + \sum_{i=1}^n \min(g_i(x) - z, 0) \\ &\leq sz + \sum_{i \in \mathcal{J}_s} \min(g_i(x) - z, 0) \\ &\leq sz + \sum_{i \in \mathcal{J}_s} (g_i(x) - z) \leq \sum_{i \in \mathcal{J}_s} g_i(x) \leq \max_x \sum_{i \in \mathcal{J}_s} g_i(x). \end{aligned}$$

Thus, we have $v(\mathcal{P}_{\text{aux}}) \leq \max_x \sum_{i \in I_s} g_i(x)$, which completes our proof for Lemma 6. $\square$

Note that Lemma 6 can also be generalized from the results given by Ogryczak and Tamir (2003). In problem $(\mathcal{P}_{\text{ellp}}(\eta, \mu))$, we first minimize the objective function (29) with respect to x by using Lemma 5, which yields the following problem:

$$(\mathcal{P}_{\text{ellp}}(\eta, \mu)): \quad \max_{H > 0, h, \xi} \quad \frac{1}{2} \sum_{i \in \mathcal{F}_s} h_i^2 / H_i + \xi, \quad (30)$$

subject to $\{H, h, \xi\}$ satisfies (28),

where $\mathcal{F}_s$ is the index set of the first s smallest elements of $\{-h_i^2 / 2H_i\}_{i=1}^n$. The index set $\mathcal{F}_s$ varies for different H and h . However, using Lemma 6 with respect to $\{-h_i^2 / H_i\}_{i=1}^n$ gives rise to the following formulation:

$$(\mathcal{P}_{\text{ellp}}(\eta, \mu)): \quad \max_{H, h, \xi, b_i, z} \quad sz + \sum_{i=1}^n b_i + \xi, \quad (31)$$

subject to $\{H, h, \xi\}$ satisfies (28),

$$\begin{pmatrix} -z - b_i & h_i \\ h_i & 2H_i \end{pmatrix} \succeq 0, \quad i = 1, \dots, n, \quad (32)$$

$$H_i > \epsilon I, \quad i = 1, \dots, n, \quad (33)$$

$$b_i \leq 0, \quad i = 1, \dots, n,$$

where ϵ is a small positive number, and $b_i, i = 1, \dots, n$, and z are auxiliary variables. Note that constraint (32) results from the inequality $b_i \leq -h_i^2 / (2H_i) - z$, for $i = 1, \dots, n$ and the Schur's complement theorem.

At this stage, the bound $(\mathcal{P}_{\text{ellp}}(\eta, \mu))$ is derived for fixed η and μ . The bound $(\mathcal{P}_{\text{ellp}}(\eta, \mu))$ can be strengthened further by the following dual search:

$$(\mathcal{P}_{\text{ellp}}): \quad \max_{\eta \geq 0, \mu \geq 0} \quad v(\mathcal{P}_{\text{ellp}}(\eta, \mu)). \quad (34)$$

From (28), we know that problem $(\mathcal{P}_{\text{ellp}})$ remains as an SDP problem even after we deal with η and μ as decision variables in problem (31).

REMARK 1. Due to constraint (28), solving the SDP problem $(\mathcal{P}_{\text{ellp}})$ is still expensive in the branch-and-bound algorithm. However, if we can specify some feasible matrix H , the SDP problem $(\mathcal{P}_{\text{ellp}})$ degenerates to the second-order cone programming problem (SCOP), which can be solved numerically faster than an SDP problem. For fixed H , if $D - H > 0$, the condition (28) can be written as

$$2c - 2\xi - (d - h)'(D - H)^{-1}(d - h) \geq 0$$

$$\implies \left\| \begin{pmatrix} 2(D - H)^{-1/2}(d - h) \\ 1 - 2(c - \xi) \end{pmatrix} \right\| \leq 1 + 2(c - \xi).$$

Similarly, we can rewrite the constraint (32) as an SOCP formulation. Thus, problem $(\mathcal{P}_{\text{ellp}})$ can be finally written as

$$(\mathcal{P}_{\text{socp}}): \quad \max_{h, \xi, b_i, z, \eta, \mu} \quad sz + \sum_{i=1}^n b_i + \xi,$$

subject to $\left\| \begin{pmatrix} 2(D - H)^{-1/2}(d - h) \\ 1 - 2(c - \xi) \end{pmatrix} \right\| \leq 1 + 2(c - \xi)$

$$\left\| \begin{pmatrix} 2H_i^{-1/2} \\ 1 + 2(z + b_i) \end{pmatrix} \right\| \leq 1 - 2(z + b_i), \quad i = 1, \dots, n,$$

$$H_i > \epsilon I, \quad i = 1, \dots, n,$$

$$b_i \leq 0, \quad i = 1, \dots, n. \quad (35)$$

Note that for a pre-fixed H , if $D - H \succeq 0$, problem $(\mathcal{P}_{\text{ellp}})$ can still be written as an SOCP formulation. We have only to use the eigenvalue decomposition to decompose matrix $D - H$ into two parts, which correspond to the nonzero eigenvalues and zero eigenvalues, respectively.

REMARK 2. Generally speaking, the bound $(\mathcal{P}_{\text{box}})$ has no dominance relationship with bound $(\mathcal{P}_{\text{ball}})$ and $(\mathcal{P}_{\text{ellp}})$. However, as a ball is just a special case of the ellipsoid, the ellipsoid bound provides more freedom and thus gives a better bound.

LEMMA 7. The bound $(\mathcal{P}_{\text{ellp}})$ is always better than the bound $(\mathcal{P}_{\text{ball}})$, i.e., $v(\mathcal{P}_{\text{ccmv}}) \geq v(\mathcal{P}_{\text{ellp}}) \geq v(\mathcal{P}_{\text{ball}})$.

PROOF. Please see the proof in the e-companion. $\square$

REMARK 3. In problem $(\mathcal{P}_{\text{ccmv}})$, the return constraint (2) and budget constraint (3) can be also formulated as inequality constraints, i.e., $r'x \geq \bar{r}$ and $\mathbf{1}'x \leq 1$. To construct the lower bound in such cases, we can still adopt the dual search schemes proposed above for $(\mathcal{P}_{\text{box}})$, $(\mathcal{P}_{\text{ball}})$, $(\mathcal{P}_{\text{ellp}})$ and $(\mathcal{P}_{\text{socp}})$, except we need to modify the searching region of dual variables to $\mu \geq 0$ and $\eta \geq 0$.

3.5. CCMV Problem Formulation with No-Shorting Constraint

The lower bounding schemes developed in the previous sections can be extended to the following CCMV problem with a no-shorting constraint:

$$(\mathcal{P}_{\text{ccmv}}^{\text{ns}}): \quad \min_x \quad \frac{1}{2} x' D x,$$

subject to $r'x \geq \bar{r}, \mathbf{1}'x \leq 1, \quad x \in \Delta(s),$

$$x \in \mathbb{R}_+^n. \quad (36)$$

Similar to problem $(\mathcal{P}_{\text{ccqo}}^p(D, d, c))$, we formulate the following relaxation problem by introducing variable $\eta \geq 0$ and $\mu \geq 0$:

$$(\mathcal{P}_{\text{ccqo}}^{p-\text{ns}}(D, d, c)) \quad \min \quad L^{\text{ns}}(x, \eta, \mu)$$

$$\triangleq \frac{1}{2} x' D x + d'x + c, \quad (37)$$

subject to $x \in \Delta(s), x \in \mathbb{R}_+^n,$ (38)

where d and c are defined in (6). Obviously, we have $v(\mathcal{P}_{\text{ccqo}}^{p-ns}(D, d, c)) \leq v(\mathcal{P}_{\text{ccmv}}^{ns})$. The method of constructing the ellipsoid bound in §3.4 can be extended to problem $(\mathcal{P}_{\text{ccmv}}^{ns})$ with some modifications. Adopting the same idea in §3.4, we consider the following class of problems:

$$(\mathcal{P}_{\text{ellp}}^{ns}(H, h, \xi)): \quad \min_x \{f_e^{ns}(x) = \frac{1}{2}x'Hx + h'x + \xi: \\ x \in \Delta(s), x \in \mathbb{R}_+^n\},$$

where H , h and ξ are defined the same as for problem (27). Note that problem $(\mathcal{P}_{\text{ellp}}^{ns}(H, h, \xi))$ can be solved explicitly for any fixed $\{H, h, \xi\}$.

LEMMA 8. Let $\bar{\mathcal{J}}_s$ be the index set of the first s smallest elements of $\{g_i^+\}_{i=1}^n$, where $g_i^+ = \max_{\xi_i \geq 0} -(h_i - \xi_i)^2 / (2H_i)$, $i = 1, \dots, n$, then the optimal value of problem $(\mathcal{P}_{\text{ellp}}^{ns}(H, h, \xi))$ is given by

$$v(\mathcal{P}_{\text{ellp}}^{ns}(H, h, \xi)) = \sum_{i \in \bar{\mathcal{J}}_s} g_i^+ + \xi.$$

PROOF. Note that minimizing $\{\frac{1}{2}H_i x_i^2 + h_i x_i\}$ for $x_i \geq 0$ yields its optimal value,

$$g_i^+ = \begin{cases} -h_i^2 / (2H_i) & \text{with } x_i^* = h_i / H_i \quad \text{if } h_i \leq 0, \\ 0 & \text{with } x_i^* = 0, \quad \text{if } h_i \geq 0. \end{cases}$$

Equivalently, g_i^+ can be also expressed as $g_i^+ = \max_{\xi_i \geq 0} -(h_i - \xi_i)^2 / (2H_i)$ by introducing auxiliary Lagrangian variables $\{\xi_i \geq 0\}_{i=1}^n$. As x should be confined in the set $\Delta(s)$, we should make only the first s smallest elements of $\{g_i^+\}_{i=1}^n$ nonzero. $\square$

To guarantee $v(\mathcal{P}_{\text{ellp}}^{ns}(H, h, \xi)) \leq v(\mathcal{P}_{\text{ccqo}}^{p-ns}(D, d, c))$, we can impose the condition similar to (28). However, since $x \in \mathbb{R}_+^n$, such a condition can be further strengthened. In particular, we should ensure $L^{ns}(x, \eta, \mu) - f_e^{ns}(x) = \frac{1}{2}x'(D - H)x + (d - h)'x + c - \xi \geq 0$ only for $x \in \{x \in \Delta(s), x \in \mathbb{R}_+^n\}$, instead of having condition (28) for entire $x \in \mathbb{R}^n$. However, the set $\{x \in \Delta(s), x \in \mathbb{R}_+^n\}$ is highly non-convex. We thus relax such a set to some convex polyhedral as follows. Let $w = (w_1, \dots, w_n)'$ such that $x_i \leq 1/w_i$ for all $i = 1, \dots, n$, i.e., w is a vector of the inverse of the upper bound of x_i . We then have the following relationship:

$$\{x \in \Delta(s), x \in \mathbb{R}_+^n\} \subset \{w'x \leq s, x \in \mathbb{R}_+^n\} \subset \mathbb{R}^n.$$

Introducing Lagrangian multipliers $\phi \in \mathbb{R}_+$ and $\beta \in \mathbb{R}_+^n$, we form the following Lagrangian function:

$$\Phi = \frac{1}{2}x'(D - H)x + (d - h)'x + c - \xi + \phi(w'x - s) - \beta'x.$$

Since Φ is a lower bound of $L^{ns}(x, \eta, \mu) - f_e^{ns}(x)$ for any $\phi \in \mathbb{R}_+$ and $\beta \in \mathbb{R}_+^n$, a sufficient condition is $\Phi \geq 0$ for all x . Thus, we have the following sufficient condition to

guarantee $v(\mathcal{P}_{\text{ellp}}^{ns}(H, h, \xi)) \leq v(\mathcal{P}_{\text{ccqo}}^{p-ns}(D, d, c))$ for all $x \in \{w'x \leq s, x \in \mathbb{R}_+^n\}$:

$$\begin{pmatrix} D - H & d - h + \phi w - \beta \\ (d - h + \phi w - \beta)' & 2(c - \xi - \phi s) \end{pmatrix} \succeq 0. \quad (39)$$

Minimizing the gap between $(\mathcal{P}_{\text{ellp}}^{ns}(H, h, \xi))$ and $(\mathcal{P}_{\text{ccqo}}^{p-ns}(D, d, c))$ yields the following lower bound problem:

$$(\mathcal{P}_{\text{ellp}}^{ns}(\eta, \mu)): \quad \max_{H \succ 0, h, \xi, \phi, \beta} \min_{x \in \mathbb{R}_+^n} f_e(x) = \frac{1}{2}x'Hx + h'x + \xi, \\ \text{subject to } x \in \Delta(s), \\ \{H, h, \xi, \phi, \beta\} \text{ satisfies (39),} \\ \phi \geq 0, \beta \in \mathbb{R}_+^n.$$

By using Lemma 8, we can solve the inner minimization problem first with respect to x , thus simplifying problem $(\mathcal{P}_{\text{ellp}}^{ns}(\eta, \mu))$ to

$$(\mathcal{P}_{\text{ellp}}^{ns}(\eta, \mu)): \quad \max_{H, h, \xi, \phi, \beta} \sum_{i \in \bar{\mathcal{J}}_s} g_i^+, \\ \text{subject to } \{H, h, \xi, \phi, \beta\} \text{ satisfies (39),} \\ \xi_i \geq 0, \quad i = 1, \dots, n, \\ \phi \geq 0, \beta \in \mathbb{R}_+^n,$$

where g_i^+ and ξ_i , $i = 1, \dots, n$, are defined in Lemma 8. By using Lemma 6, we can further rewrite $(\mathcal{P}_{\text{ellp}}^{ns}(\eta, \mu))$ as an SDP formulation:

$$(\mathcal{P}_{\text{ellp}}^{ns}): \quad \max_{H, h, \xi, b_i, z, \phi, \beta, \eta, \mu} sz + \sum_{i=1}^n b_i + \xi, \\ \text{subject to } \{H, h, \xi, \phi, \beta\} \text{ satisfies (39),} \\ \begin{pmatrix} -z - b_i & h_i - \xi_i \\ h_i - \xi_i & 2H_i \end{pmatrix} \succeq 0, \\ i = 1, \dots, n, \\ H_i \succ \epsilon I, \quad i = 1, \dots, n, \\ b_i \leq 0, \xi_i \geq 0, \quad i = 1, \dots, n, \\ \phi \geq 0, \beta \in \mathbb{R}_+^n, \eta \geq 0, \mu \geq 0,$$

which yields an ellipsoid bound. Note that in problem $(\mathcal{P}_{\text{ellp}}^{ns})$ we take η and μ as the decision variables, thus completing the dual search in one shot. Problem $(\mathcal{P}_{\text{ellp}}^{ns})$ can be also reformulated as an SOCP problem, $(\mathcal{P}_{\text{socp}}^{ns})$, by using the same argument provided in Remark 1, if some feasible matrix $H \succeq 0$ can be specified.

REMARK 4. Note that the box and ball bounds developed in §3.2 can be directly extended for problem $(\mathcal{P}_{\text{ccqo}}^{p-ns}(D, d, c))$ by introducing n Lagrangian variables for the constraint $x \in \mathbb{R}_+^n$. However, in the dual search, we have to search for $n + 2$ variables, which might cause inefficiency in real implementation.

4. Sharpe Ratio Maximization and Global Variance Minimization Problems

If a risk free asset is included in the portfolio, the CCMV problem then becomes the following cardinality constrained Sharpe ratio maximization problem:

$$(\mathcal{P}_{\text{ccSharpe}}): \quad \min_{x \in \Delta(s)} \quad \frac{1}{2} x' D x, \\ \text{subject to} \quad r' x + r_f x_f = \bar{r}, \\ \mathbf{1}' x + x_f = 1,$$

where x_f and r_f are the portfolio weight and the (deterministic) return of the risk free asset, respectively. Note that problem $(\mathcal{P}_{\text{ccSharpe}})$ generates the largest Sharpe ratio from among all the portfolios with the cardinality constraint of at most s risky assets.

Recently, the global minimum-variance portfolio has been attracting increasing attention due to its robustness in portfolio weight (DeMiguel and Nogales 2009). Thus, we also consider in this research the following cardinality constrained global minimum variance problem:

$$(\mathcal{P}_{\text{cc-minvar}}): \quad \min_{x \in \Delta(s)} \quad \frac{1}{2} x' D x, \\ \text{subject to} \quad \mathbf{1}' x = 1.$$

While problems $(\mathcal{P}_{\text{ccSharpe}})$ and $(\mathcal{P}_{\text{cc-minvar}})$ have important applications in financial practice, we will show in this section that these two problems are essentially equivalent to the pure CCQO problem $(\mathcal{P}_{\text{ccqo}}^p)$ defined in §2. More specifically, we prove now that problems $(\mathcal{P}_{\text{ccSharpe}})$ and $(\mathcal{P}_{\text{cc-minvar}})$ can be solved once the solution of problem $(\mathcal{P}_{\text{ccqo}}^p)$ is known.

THEOREM 5. *Let x_{ccqo}^p , x_{ccSharpe} and $x_{\text{cc-minvar}}$ be the solutions to problems $(\mathcal{P}_{\text{ccqo}}^p)$, $(\mathcal{P}_{\text{ccSharpe}})$ and $(\mathcal{P}_{\text{cc-minvar}})$, respectively. If we set $\bar{d} = r - r_f \mathbf{1}$ and $\bar{c} = 0$ in problem $(\mathcal{P}_{\text{ccqo}}^p)$, then there exists $\beta_1 \in \mathbb{R}$ such that $x_{\text{ccSharpe}} = \beta_1 x_{\text{ccqo}}^p$ with $v(\mathcal{P}_{\text{ccSharpe}}) = -(\bar{r} - r_f)^2 / (4v(\mathcal{P}_{\text{ccqo}}^p))$. If we set $\bar{d} = -\mathbf{1}$ and $\bar{c} = 0$ in problem $(\mathcal{P}_{\text{ccqo}}^p)$, then there exists $\beta_2 \in \mathbb{R}$ such that $x_{\text{cc-minvar}} = \beta_2 x_{\text{ccqo}}^p$ with $v(\mathcal{P}_{\text{cc-minvar}}) = -1 / (4v(\mathcal{P}_{\text{ccqo}}^p))$.*

PROOF. Please see the proof in the e-companion. $\square$

The most important conclusion from Theorem 5 is that both problems $(\mathcal{P}_{\text{ccSharpe}})$ and $(\mathcal{P}_{\text{cc-minvar}})$ have the same optimal sparsity as problem $(\mathcal{P}_{\text{ccqo}}^p)$ if $\bar{d}$ and $\bar{c}$ are specified with the particular values given in Theorem 5. After determining the optimal sparsity from solving the pure cardinality constrained problem $(\mathcal{P}_{\text{ccqo}}^p)$, problem $(\mathcal{P}_{\text{ccSharpe}})$ or $(\mathcal{P}_{\text{cc-minvar}})$ can easily be solved as a convex optimization problem corresponding to this identified optimal sparsity.

Recall that for general CCMV problem $(\mathcal{P}_{\text{ccmv}})$, the solution method involves two phases, (i) constructing lower bounds for $(\mathcal{P}_{\text{ccqo}}^p(D, d(\eta, \mu), c(\eta, \mu)))$, the Lagrangian relaxation of $(\mathcal{P}_{\text{ccmv}})$, and (ii) performing dual search with respect to Lagrangian multipliers η and μ . The special

structures of $(\mathcal{P}_{\text{ccSharpe}})$ and $(\mathcal{P}_{\text{cc-minvar}})$, i.e., their equivalence to the pure cardinality constrained quadratic optimization problem, help us remove the need of dual search. From the algorithm point of view, we need only to invoke phase (i) of the solution method for $(\mathcal{P}_{\text{ccqo}}^p(D, \bar{d}, \bar{c}))$ with fixed $\bar{d}$ and $\bar{c}$ when dealing with $(\mathcal{P}_{\text{ccSharpe}})$ and $(\mathcal{P}_{\text{cc-minvar}})$, thus improving further the efficiency of the computational algorithms proposed earlier for $(\mathcal{P}_{\text{ccmv}})$.

5. Heuristics for Identifying Good Feasible Solution

In the previous subsections, we have developed various lower bounding schemes for the CCMV problem. From the results of the numerical tests (see §6), we find that the ellipsoid bounds, $(\mathcal{P}_{\text{ellp}})$ and $(\mathcal{P}_{\text{ellp}}^{ns})$, are extremely tight for the CCMV problem without or with no-shorting constraint, respectively. We demonstrate in this subsection that such lower bounds not only provide very good approximation of the optimal value but also supply useful information to help identify a good feasible solution for the CCMV problem.

As an example, we illustrate our heuristics on how to identify a good feasible solution to problem $(\mathcal{P}_{\text{ccmv}})$ from solving problem $(\mathcal{P}_{\text{ellp}})$. Note that solving the CCMV problem is basically equivalent to identifying the optimal sparsity $\{\delta(x_1), \dots, \delta(x_n)\}$ of the decision vector x , since once the sparsity is known we can solve the convex quadratic programming problem easily according to this specific sparsity. Suppose that $\{b_i^*\}_{i=1}^n$ solves $(\mathcal{P}_{\text{ellp}})$, where b_i , $i = 1, \dots, n$, are the auxiliary variables defined in problem $(\mathcal{P}_{\text{ellp}})$. Let $\mathcal{J}_b$ be the index set of the first s largest elements of $\{|b_i^*|\}_{i=1}^n$. Then, we can fix the sparsity $z = (z_1, \dots, z_n)' \in \{0, 1\}^n$ according to $\mathcal{J}_b$ as follows: Let $z_i = 1$ for $i \in \mathcal{J}_b$, and $z_i = 0$ for $i \notin \mathcal{J}_b$. Solving problem $(\mathcal{P}_{\text{ccmv}})$ according to this sparsity z yields a feasible solution of problem $(\mathcal{P}_{\text{ccmv}})$, i.e., fix $x_i = 0$ for $z_i = 0$, and let x_i be a free decision variable for $z_i = 1$ in problem $(\mathcal{P}_{\text{ccmv}})$. We denote such a problem as $(\mathcal{P}_{\text{ccmv}}(z))$ ($\mathcal{P}_{\text{ccmv}}^{ns}(z)$ for the no-shorting case). Note that when $s \geq 2$, problem $(\mathcal{P}_{\text{ccmv}}(z))$ always has solution, since the number of free decision variables is s and the number of equality constraint is 2. For both the problems $(\mathcal{P}_{\text{ccSharpe}})$ and $(\mathcal{P}_{\text{cc-minvar}})$, since they are equivalent to the pure CCQO problem $(\mathcal{P}_{\text{ccqo}}^p)$, we can implement the above heuristics to find the sparsity of a good feasible solution for the corresponding problem $(\mathcal{P}_{\text{ccqo}}^p)$ and then use such a sparsity to identify a feasible solution of problem $(\mathcal{P}_{\text{ccSharpe}})$ or $(\mathcal{P}_{\text{cc-minvar}})$.

For the problem with the no-shorting constraint, $(\mathcal{P}_{\text{ccmv}}^{ns})$, we can use the same procedure to get the sparsity z from the solution of problem $(\mathcal{P}_{\text{ellp}}^{ns})$. However, there are two different cases.

(i) If there exists $i^* \in \mathcal{J}_b$ such that $r_{i^*} \geq \bar{r}$, then problem $(\mathcal{P}_{\text{ccmv}}^{ns}(z))$ is a convex quadratic programming problem and possesses a feasible solution. The feasibility follows from the fact that the feasible region of problem $(\mathcal{P}_{\text{ccmv}}^{ns}(z))$ is

$$\left\{ x \mid \sum_{i \in \mathcal{J}_b} x_i r_i \geq \bar{r}, \sum_{i \in \mathcal{J}_b} x_i \leq 1, x_i \geq 0, i \in \mathcal{J}_b, x_i = 0, i \notin \mathcal{J}_b \right\}.$$

(ii) If there exists no $i^* \in \mathcal{J}_b$ such that $r_{i^*} \geq \bar{r}$, then $(\mathcal{P}_{\text{ccmv}}^{\text{ns}})(z)$ is an infeasible problem. For this case, we can replace one index $i_1 \in \mathcal{J}_b$ by some index $i_2 \in \bar{\mathcal{J}}_b$, where $\bar{\mathcal{J}}_b$ is the complementary set of $\mathcal{J}_b$, such that the problem goes back to Case (i). In our algorithm, we remove i_1 out of $\mathcal{J}_b$, where $i_1 = \arg \min_{i \in \mathcal{J}_b} |b_i|$ and we choose index $i_2 \in \bar{\mathcal{J}}_b$ as the entering index, where the indices in $\bar{\mathcal{J}}_b$ are arranged according to the descending order of their $|b_i|$ and i_2 is the first index that satisfies $r_i \geq \bar{r}$ with $i \in \bar{\mathcal{J}}_b$. Note that such a procedure takes at most $O(n)$ time to find suitable i_1 and i_2 in order to change the problem back to Case (i).

Theoretically evaluating the quality of the feasible solution generated from the above procedure is very difficult. From our numerical tests, however, such a feasible solution provides a good approximation of the real optimal solution. Furthermore, for pure CCOP formulation $(\mathcal{P}_{\text{ccqo}}^p)$, if $\bar{d}$, $\bar{c}$, and matrix D possess certain special structure, then the theoretical evaluation of such a feasible solution can also be obtained.

THEOREM 6. *In problem $(\mathcal{P}_{\text{ccqo}}^p)$ defined in (4), if $\bar{d} = \mathbf{1}$, $\bar{c} = 0$ and D can be decomposed to $D = A + U$, where*
(i) $U = \text{diag}\{u_1, \dots, u_n\}$ with $u_i > 0$ for $i = 1, \dots, n$, and
(ii) $A \geq 0$ and $\|A\| \leq \epsilon \kappa$, where $\epsilon > 0$, $\kappa \triangleq \lambda_{\min}^D \theta_1 / \theta_s$ with θ_i and $\lambda_{\min}^D$ being the i th smallest element of $\{u_i\}_{i=1}^n$ and the smallest eigenvalue of D , respectively, then we have the following approximation ratio:

$$\frac{|v(\mathcal{P}_{\text{ellp}}) - v(\mathcal{P}_{\text{ccqo}}^p)|}{|v(\mathcal{P}_{\text{ellp}})|} \leq \epsilon. \quad (40)$$

PROOF. Please see the proof in the e-companion. $\square$

Note that when constructing the feasible solution in the proof of Theorem 6, we set $\delta(x_i) = 1$, for $i = 1, \dots, s$ and $\delta(x_i) = 0$ for $i = s + 1, \dots, n$ according to the magnitude of $|b_i|$, $i = 1, \dots, n$, as $b_i = -z - 1/(2u_i)$ for $i = 1, \dots, s$ and $b_i = 0$ for $i = s + 1, \dots, n$. This strategy is thus the same as the heuristic one that we propose in the beginning of this subsection.

Theorem 6 basically states that when matrix D is not deviating too much from a diagonal matrix U , the lower bound $(\mathcal{P}_{\text{ellp}})$ is very tight. As we showed in the previous section, problem $(\mathcal{P}_{\text{cc-minvar}})$ is closely related to problem $(\mathcal{P}_{\text{ccqo}}^p)$ when $\bar{d} = -\mathbf{1}$ and $\bar{c} = 0$. Such a fact partially explains why both the lower bound and the feasible solution from $(\mathcal{P}_{\text{ellp}})$ are of very high quality for the minimum-variance portfolio selection problem $(\mathcal{P}_{\text{cc-minvar}})$ (see Table 3 in §6). Although matrix D (related to covariance matrix) is not diagonally dominant in general, the variance term of each stock is usually much larger than the covariance between this stock and the others. Especially when the shrinkage estimator is used, the elements on the off-diagonal can be significantly reduced (see Ledoit and Wolf 2003).

6. Computational Results

6.1. Problem Generation

To evaluate the solution schemes proposed in §§3–5, we construct various test problems by using the historical stock data from the CRSP database provided by the Wharton Research Database Service (WRDS). In particular, based on the constituent stocks of Standard & Poor's 500 (Large Capital) and Standard & Poor's MidCap 400, we choose 720 stocks to form the stock pool and use their monthly return data between January 1979 and December 2009 to estimate the mean and covariance matrix. Among these 720 stocks, some have the full observations with a sample size of 372 (Jan. 1979–Dec. 2009), while some have only partial observations, due to the late starting time when they began to be included in the index. Thus, when the size of a test problem is large, we will choose these stocks with full observations in order to get good estimation of the mean and covariance matrix. We focus mainly on testing the various lower bounds for problems $(\mathcal{P}_{\text{ccmv}})$, $(\mathcal{P}_{\text{ccmv}}^{\text{ns}})$, $(\mathcal{P}_{\text{ccSharpe}})$ and $(\mathcal{P}_{\text{cc-minvar}})$. The testing problems of different sizes are generated by randomly selecting n stocks from the stock pool. We use the shrinkage estimator to estimate the covariance matrix. The shrinkage estimator is a balanced summation of the sample covariance matrix and the covariance matrix estimated from the factor model, which has various advantages over the sample covariance matrix (see Ledoit and Wolf 2003 for a detailed discussion). For simplicity, the mean vector of the stock returns is estimated by the sample mean. Although, as pointed out by several papers in the literature (DeMiguel et al. 2007, Goldfarb and Iyengar 2003), estimating the mean of the return by using only historical data is not stable, we still decide to adopt such a simple way to estimate the mean as our purpose is only to test the efficiency of our algorithm.

All the problems are solved under the Linux platform (2.8 g CPU, 4 Gb RAM). In particular, the quadratic programming problem is solved by calling CPLEX.12.3 (2011), the semidefinite programming problem is solved by using SDPA 7.3.1 (see Yamashita et al. 2003), and the second-order cone programming problem is solved by using MOSEK (2012). Furthermore, the lower bounds $v(\mathcal{P}_{\text{box}})$ and $v(\mathcal{P}_{\text{ball}})$ are coded in C++ with LAPACK.

We have tested various lower bounds developed in §3 and have compared them with the existing lower bounds suggested in the literature. Due to the page limit, we have placed this comparison result in the e-companion to this paper.

6.2. Performance Comparison of Branch and Bound Algorithms

In this subsection, we report how to integrate our lower bounding schemes into a branch-and-bound (BnB) algorithm to identify exact solutions to problems $(\mathcal{P}_{\text{ccmv}})$, $(\mathcal{P}_{\text{ccSharpe}})$, $(\mathcal{P}_{\text{cc-minvar}})$, and $(\mathcal{P}_{\text{ccmv}}^{\text{ns}})$. The general mechanism

of our branch-and-bound method can be stated as follows. We gradually fix the sparsity $\delta(x_i)$ at either 1 or 0 according to some order. When total s of $\delta(x_i)$ s are fixed at 1, we obtain a feasible solution, and the incumbent is updated when the new feasible solution is better. Once one additional $\delta(x_i)$ is fixed at 0, a lower bound of this branch is computed. If this lower bound is larger than the objective value of the incumbent, this branch is fathomed. This computational procedure easily can be organized as a binary search tree.

In our BnB algorithm, we adopt a hybrid lower bounding scheme. As the continuous relaxation ($\mathcal{P}_{\text{cnt}}$) is the cheapest to solve, we calculate the continuous bound first at each node of the binary search tree. If the continuous bound $v(\mathcal{P}_{\text{cnt}})$ fails to prune the corresponding branch, we consider other relaxations, such as ($\mathcal{P}_{\text{box}}$) and ($\mathcal{P}_{\text{socp}}$). As shown in Figures EC.3 and EC.4, the high computational cost of the bounds $v(\mathcal{P}_{\text{ellp}})$ and $v(\mathcal{P}_{\text{socp}})$ prevents us from using these bounds fully for all the nodes in the branch-and-bound algorithm. Thus, for problems ($\mathcal{P}_{\text{ccmv}}$), ($\mathcal{P}_{\text{ccSharpe}}$), ($\mathcal{P}_{\text{cc-minvar}}$) and ($\mathcal{P}_{\text{ccmv}}^{\text{ns}}$), the tight bounds of $v(\mathcal{P}_{\text{ellp}})$ and $v(\mathcal{P}_{\text{socp}})$ are computed only for some selected nodes according to the following strategy. We define the *depth* for any node as the number of the sparsity $\delta(x_i)$ fixed at 0. Then the bound $v(\mathcal{P}_{\text{socp}})$ is computed when the *depth* of the nodes is smaller than a given level, L , using the dual search scheme developed in §3.4. A cheap bound $v(\mathcal{P}_{\text{box}})$ (or $v(\mathcal{P}_{\text{ball}})$) is computed otherwise using the dual search scheme developed in §§3.2 and 3.3. This strategy follows from the fact that when the *depth* grows, the gap between $v(\mathcal{P}_{\text{socp}})$ and $v(\mathcal{P}_{\text{box}})$ (or $v(\mathcal{P}_{\text{ball}})$) decreases. In our numerical experiments for problems without no-shorting constraint, we set $L = 5$ when problem dimension n is less than 100, $L = 10$ when $100 \leq n \leq 120$, and $L = 15$ when $120 < n \leq 200$. For problems with no-shorting constraint, we set $L = 5$ when $n \leq 140$, and $L = 10$ when $140 < n \leq 300$. The larger the size of the problem, the higher the importance of tighter bounds generated by deep cuts. The ellipsoid bound $v(\mathcal{P}_{\text{ellp}})$ or $v(\mathcal{P}_{\text{ellp}}^{\text{ns}})$ is computed only in the root node, which provides the global lower bound J_L and the solution matrix H (see §3.4). For all the subproblems, we use the corresponding diagonal elements of matrix H as the input parameter for computing the bound $v(\mathcal{P}_{\text{socp}})$ (or $v(\mathcal{P}_{\text{socp}}^{\text{ns}})$) of the corresponding subproblems. The global lower bound J_L is used to define a stopping criteria, i.e., to define

$$\text{gap} = (\bar{J} - J_L)/\bar{J},$$

where $\bar{J}$ is the objective value of the incumbent. After computing the bound $v(\mathcal{P}_{\text{ellp}})$ or $v(\mathcal{P}_{\text{ellp}}^{\text{ns}})$ in the root node, we also get a feasible solution (see §5), which serves as the initial incumbent. In addition to the general situation where our branch-and-bound algorithm stops after implicitly visiting all nodes in the binary search tree, our algorithm may also stop when the *gap* meets the given level, 0.001 in our

algorithm. The initial branching order is fixed according to the solution of the continuous relaxation, i.e., branching the variable with the largest absolute value first. During the search process, we also use the solution of the continuous relaxation ($\mathcal{P}_{\text{cnt}}$) for the subproblem to update the branching order dynamically, i.e., to select the variable with the largest absolute value to branch first in its children nodes. For problem ($\mathcal{P}_{\text{ccmv}}^{\text{ns}}$) where the no-shorting constraint is imposed, instead of using the box bound, we use the continuous relaxation ($\mathcal{P}_{\text{cnt}}^{\text{ns}}$) as a cheap bound. Note that for problems ($\mathcal{P}_{\text{ccmv}}$) (($\mathcal{P}_{\text{ccSharpe}}$) and ($\mathcal{P}_{\text{cc-minvar}}$)) and ($\mathcal{P}_{\text{ccmv}}^{\text{ns}}$), the two cheap bounds $v(\mathcal{P}_{\text{box}})$ and $v(\mathcal{P}_{\text{cnt}}^{\text{ns}})$ are computed, respectively, by using warm-start strategy. Such a warm-start strategy follows the same idea in Bertsimas and Shioda (2009): (i) When computing the bound $v(\mathcal{P}_{\text{box}})$, the most expensive step is to compute the inverse of a submatrix of D . We then store the inverse of the given submatrix of D in the parent node and modify it for its children nodes. (ii) When computing bound $v(\mathcal{P}_{\text{cnt}}^{\text{ns}})$, we store the Lemke pivoting table in each node and modify the table according to the method in Bertsimas and Shioda (2009) to compute the lower bound in the children nodes.

We compare the performance of our BnB algorithm with the standard CPLEX solver and the BnB-cnt algorithm proposed by Bertsimas and Shioda (2009). The CPLEX solver solves the mixed-integer formulation ($\mathcal{P}_{\text{mip}}$) for all types of problems. In such a formulation, we fix the big- M as $4/s$, where s is the given cardinality of the portfolio. Clearly, a suitable magnitude of big- M is related to the target return $\bar{r}$. In our tests, we set the $\bar{r}$ as the median of the expected returns of the given pool of stocks. In our experiment, $4/s$ works well for most of the tested examples. When CPLEX reports infeasibility for some rare cases, we then use a safe choice of big- M with $M = 1$ for these cases. Our BnB algorithm and the BnB-cnt algorithm of Bertsimas and Shioda (2009) adopt similar warm-start strategies. However, a hybrid lower-bounding scheme is used in our BnB algorithm, while only the continuous relaxation is used in the BnB-cnt algorithm. We set the stopping criteria of gap ratio as 0.001 for all the solvers in our tests. We also set an upper bound of execution time for all the solvers as half an hour (1,800 seconds). In problem ($\mathcal{P}_{\text{ccSharpe}}$), we set the annual risk free rate as 2%, thus $r_f = 1 + 0.02/12$.

The computational results are listed in Tables 1–4, where the columns “Node,” “Time,” and “Suc” record the number of nodes explored, CPU time in seconds, and the number of problems being solved successfully (in total 10 problems), respectively. The column “ ψ ” denotes the *improvement ratio*, i.e.,

$$\psi \triangleq (\bar{J} - J^*)/\bar{J}, \quad (41)$$

where J^* is the objective value returned from a BnB solver in a given time limit and $\bar{J}$ is the objective value of a good feasible solution generated using the method illustrated in §5. The value of ψ measures the improvement of

Table 1. Comparison of CPLEX, BnB-cnt, and BnB solvers for problem ($\mathcal{P}_{\text{ccmv}}$).

n	s	CPLEX			BnB-cnt			BnB				L
		Nodes (10^3)	Time (sec)	Suc	Nodes (10^3)	Time (sec)	Suc	Nodes (10^3)	Time (sec)	Suc	ψ (%)	
60	5	41.8	10.1	10	21.4	0.8	10	0.0	1.3	10	1.80	5
60	10	1,772.7	310.4	10	521.4	23.2	10	0.1	1.4	10	0.66	5
80	5	178.7	60.4	10	157.1	17.4	10	4.8	6.3	10	4.85	5
80	10	6,943.3	1,750.1	2	9,655.6	1,137.1	8	249.3	69.6	10	1.07	5
100	5	492.2	190.7	10	384.5	126.7	10	3.1	26.7	10	2.64	10
100	10	6,178.2	1,800.0	0	9,953.9	1,760.0	1	113.1	144.7	10	1.00	10
100	15	6,081.7	1,800.0	0	8,401.8	1,800.0	0	21.9	41.4	10	1.08	10
120	5	901.8	400.1	10	1,126.7	321.3	10	4.8	28.6	10	10.70	10
120	10	4,646.6	1,800.0	1	5,774.2	1,789.0	1	432.0	348.1	10	2.69	10
120	15	5,331.0	1,800.0	0	6,263.4	1,800.0	0	1.9	74.3	10	0.47	10
140	10	3,846.3	1,800.0	0	1,121.3	1,800.0	0	234.2	757.9	7	1.95	15
140	15	4,193.8	1,800.0	0	1,422.2	1,800.0	0	81.7	581.3	8	1.29	15
200	15	1,642.2	1,800.0	0	2,921.4	1,800.0	0	325.3	1,653.5	1	0.92	15
200	20	1,760.9	1,800.0	0	2,990.8	1,800.0	0	305.5	1,721.1	1	0.74	15

Table 2. Comparison of CPLEX, BnB-cnt, and BnB solvers for problem ($\mathcal{P}_{\text{ccSharpe}}$).

n	s	CPLEX			BnB-cnt			BnB				L
		Nodes (10^3)	Time (sec)	Suc	Nodes (10^3)	Time (sec)	Suc	Nodes (10^3)	Time (sec)	Suc	ψ (%)	
60	5	53.2	13.1	10	65.6	1.1	10	2.7	2.4	10	1.02	5
60	10	4,548.3	1,097.4	9	6,253.7	265.4	10	1.3	1.7	10	0.70	5
80	5	274.5	95.4	10	349.4	55.5	10	4.3	4.7	10	1.70	5
80	10	6,608.3	1,780	1	12,144.3	1,519.4	4	88.4	15.2	10	0.45	5
100	5	773.7	316.7	10	1,284.8	379.4	10	20.1	61.0	10	2.68	10
100	10	5,737.6	1,800	0	5,860.8	1,800.0	0	5.5	38.9	10	0.54	10
100	15	6,288.6	1,800	0	5,788.5	1,800.0	0	7.8	77.8	10	0.59	10
120	5	1,238.7	620.1	10	3,364.3	1,263.0	9	649.9	272.9	10	2.09	10
120	10	4,797.8	1,800	0	3,948.6	1,800.0	0	915.6	449.5	10	1.64	10
120	15	4,618.0	1,800	0	3,847.9	1,800.0	0	802.1	574.2	7	1.35	10
140	10	3,647.8	1,800	0	3,221.1	1,800.0	0	692.1	730.3	7	1.47	15
140	15	3,704.0	1,800	0	5,008.7	1,800.0	0	824.6	1,306.0	3	0.95	15
200	15	1,882.0	1,800	0	3,325.2	1,800.0	0	616.0	1,800	0	1.49	15
200	20	1,766.9	1,800	0	4,845.6	1,800.0	0	500.7	1,622.5	1	1.37	15

Table 3. Comparison of CPLEX, BnBcnt, and BnB solvers for problem ($\mathcal{P}_{\text{cc-minvar}}$).

n	s	CPLEX			BnB-cnt			BnB				L
		Nodes (10^3)	Time (sec)	Suc	Nodes (10^3)	Time (sec)	Suc	Nodes (10^3)	Time (sec)	Suc	ψ (%)	
60	5	18.2	5.3	10	10.4	0.9	10	0.01	0.7	10	0.20	5
60	10	925.6	203.8	10	518.6	32.6	10	0.01	1.0	10	0.02	5
80	5	134.7	44.3	10	116.9	23.0	10	1.25	3.6	10	0.50	5
80	10	6,544.3	1,766.1	2	7,891.9	1,304.0	6	5.09	5.6	10	0.70	5
100	5	272.7	105.8	10	288.1	107.5	10	0.14	4.1	10	0.28	10
100	10	6,257.8	1,800.0	0	5,338.0	1,800.0	0	0.19	10.3	10	0.80	10
100	15	6,643.1	1,800.0	0	5,346.5	1,800.0	0	0.05	7.6	10	0.09	10
120	5	245.7	245.7	10	751.9	238.7	10	0.03	4.8	10	0.40	10
120	10	5,082.7	1,800.0	0	4,778.0	1,800.0	0	0.64	8.6	10	0.20	10
120	15	4,980.7	1,800.0	0	4,007.7	1,800.0	0	30.23	89.1	10	0.39	10
140	10	3,696.5	1,800.0	0	2,896.7	1,800.0	0	153.14	602.4	9	1.19	15
140	15	3,765.9	1,800.0	0	4,693.4	1,800.0	0	747.45	642.9	9	0.85	15
200	15	1,977.1	1,800.0	0	3,060.4	1,800.0	0	408.40	1,442.5	5	0.70	15
200	20	1,845.3	1,800.0	0	2,724.6	1,800.0	0	319.37	1,382.2	4	0.70	15

Table 4. Comparison of CPLEX, BnBcnt, and BnB solvers for problem $(\mathcal{P}_{ccmv}^{ns})$.

n	s	CPLEX			BnB-cnt			BnB				
		Nodes (10^3)	Time (sec)	Suc	Nodes (10^3)	Time (sec)	Suc	Nodes (10^3)	Time (sec)	Suc	ψ (%)	L
80	5	26.6	7.6	10	12.2	24.7	10	0.48	4.7	10	4.70	5
80	10	20.5	5.3	10	11.4	29.0	10	0.01	1.3	10	0.54	5
100	5	51.4	25.9	10	17.8	75.3	10	0.45	9.6	10	2.31	5
100	10	38.2	17.2	10	16.5	87.9	10	0.01	2.6	10	0.13	5
140	10	150.0	115.3	10	41.5	436.2	10	0.14	9.0	10	0.23	5
140	15	23.1	11.3	10	3.1	54.2	10	0.01	4.0	10	0.00	5
180	5	184.3	103.7	10	57.3	1,535.4	7	4.24	170.3	10	6.91	10
180	10	497.4	372.1	10	48.2	1,641.0	8	0.02	14.6	10	0.49	10
200	5	271.9	218.0	10	33.5	1,321.7	5	4.23	276.4	9	3.40	10
200	10	946.5	623.0	7	25.6	1,194.7	6	0.63	176.4	10	0.69	10
240	10	947.2	1,393.4	8	18.9	1,665.4	1	0.09	62.0	10	0.64	10
240	15	393.5	540.6	10	94.2	942.3	6	0.02	21.0	10	0.00	10
260	10	1,069.6	971.9	6	13.7	1,626.8	2	0.82	212.3	9	0.44	10
260	15	605.7	458.1	10	8.8	1,120.4	4	0.01	24.7	10	0.13	10
300	15	143.3	137.1	10	2.4	487.4	10	0.02	45.6	10	0.00	10
300	20	145.0	388.0	10	0.8	1,671.0	4	0.02	13.6	10	0.00	10

the solution found by a BnB solver when compared with $\bar{J}$. The column “ L ” is the upper bound of the *depth* under which the bound $v(\mathcal{P}_{socc})$ or $v(\mathcal{P}_{socc}^{ns})$ is carried out as the deep cut.

We observe from Table 1 that our BnB algorithm solves all the cases successfully for problem types 1–10, while the CPLEX solver and the BnB-cnt algorithm (Bertsimas and Shioda 2009) solve only part of them. Compared with the CPLEX solver and the BnB-cnt algorithm, our BnB algorithm consumes significantly less CPU time to identify the optimal solution. Furthermore, our BnB algorithm explored many fewer nodes in the whole searching procedure, due to the tightness and efficiency of our hybrid bounding strategy. For problem types 11–14, $((n = 140, s = 10), (n = 140, s = 15), (n = 200, s = 15), (n = 200, s = 20))$, our BnB algorithm solves more cases successfully than both the CPLEX solver and the BnB-cnt solver. Furthermore, our BnB solver gets a better feasible solution (portfolio), for every individual case, when comparing the average objective values (variance) returned by different solvers. The column “ ψ ” shows that the worst case $(n = 120, s = 5)$ of improvement ratio is still less than 10.7%.

Tables 2 and 3 show a similar pattern as Table 1: our BnB algorithm is the most efficient in solving problems $(\mathcal{P}_{ccSharpe})$ and $(\mathcal{P}_{cc-minvar})$ among the three solvers. This is especially true for the global variance minimization problem $(\mathcal{P}_{cc-minvar})$. Evidenced from Table 3, our BnB algorithm is able to solve most of the cases in very short CPU time (< 10 seconds) for problem types 1–9. Due to the tightness of the bounds $v(\mathcal{P}_{ellp})$ and $v(\mathcal{P}_{socc})$, most of the branches are pruned when the *depth* of active nodes is small. We can also observe in Table 3 that improvement ratio ψ for problem $(\mathcal{P}_{ccvbm})$ is even less than 1.2% for all types of problems, which further demonstrates that the heuristic feasible solution we have proposed in §5 is indeed a good approximation of the real optimal portfolio. Note that our

BnB algorithm performs better than the other solvers, not only in the sense of the average value but also in each individual case.

Table 4 shows that our BnB solver also performs better than both the CPLEX solver and the BnB-cnt algorithm in most cases for problem $(\mathcal{P}_{ccmv}^{ns})$, where the no-shorting constraint is included. Except for the problem types of $(n = 180, s = 5)$ and $(n = 200, s = 5)$, our BnB algorithm consumes the least CPU time to achieve the best portfolio. Furthermore, our BnB algorithm is able to successfully solve more cases of different types of problems than the CPLEX solver and the BnB-cnt solver in the given time limit. Based on Table 4, we have some interesting observations that make the solution pattern of problem $(\mathcal{P}_{ccmv}^{ns})$ different from those of problems $(\mathcal{P}_{ccmv})$, $(\mathcal{P}_{ccSharpe})$ and $(\mathcal{P}_{cc-minvar})$. Generally speaking, imposing the no-shorting constraint makes the CCMV problem relatively easier to solve. Note that the CPLEX solver, the BnB-cnt algorithm, and our BnB algorithm all perform better, in terms of nodes and time, for larger cardinality when fixing n , which seems to contradict our intuition that a complexity measure of the problem, C_n^s , increases with s , when $s < n/2$. We can also observe that the improvement ratio is 0 for almost all the tested cases with $s = 15$ and $s = 20$ (except one). That is to say, our heuristic feasible portfolio is actually the real optimal portfolio.

Generally speaking, the BnB-cnt algorithm proposed by Bertsimas and Shioda (2009) performs better than CPLEX for problems $(\mathcal{P}_{ccmv})$, $(\mathcal{P}_{ccSharpe})$ and $(\mathcal{P}_{cc-minvar})$, due to the efficiency of its warm-start strategy for computing the continuous relaxation. For some problems with $n \leq 60$, the BnB-cnt even performs better than our algorithm. However, when $n \geq 80$, our algorithm performs much better than the BnB-cnt algorithm.

Even in situations where none of the solvers is able to identify the exact solution, our BnB algorithm, in general,

Table 5. Comparison of CPLEX, BnB-cnt, and BnB solvers with CPU time ≤ 120 s.

	n	s	CPLEX			BnB-cnt			BnB			
			Nodes (10^3)	Suc	Obj	Nodes (10^3)	Suc	Obj	Nodes (10^3)	Suc	Obj	L
$(\mathcal{P}_{\text{ccmv}})$	300	10	48.9	0	350.89	67.0	0	364.5	5.6	0	334.2	10
	300	15	48.7	0	292.86	66.5	0	306.8	4.6	0	281.1	10
	320	10	37.0	0	351.83	65.4	0	364.2	1.7	0	335.6	10
	320	15	44.0	0	294.33	52.8	0	317.0	1.5	0	280.5	10
	350	10	28.7	0	350.21	48.5	0	362.4	0.9	0	335.8	10
	350	15	30.3	0	293.24	45.7	0	302.5	1.1	0	280.0	10
$(\mathcal{P}_{\text{ccmv}}^{\text{ns}})$	320	8	176.35	0	357.74	0.56	0	357.96	0.12	1	357.74	10
	320	10	146.97	0	333.61	0.52	0	333.99	0.13	2	333.51	10
	320	12	85.17	7	317.83	0.50	0	318.41	0.02	9	317.63	10
	350	8	164.44	0	354.63	0.37	0	354.63	0.08	0	354.63	10
	350	10	106.87	0	330.28	0.34	0	330.79	0.09	0	330.28	10
	350	12	84.70	7	315.89	0.33	0	316.25	0.02	8	315.64	10

identifies a better feasible solution than the other solvers do. Table 5 shows the comparison results of different solvers for relatively large-size problems within a short time limit (≤ 120 s). To avoid computational errors caused by very small numbers, we assume in every test that the initial wealth is \$1,000. Thus, “Obj” in Table 5 represents the variance of the portfolio with an initial wealth of \$1,000. From Table 5, we can see that our BnB algorithm is able to find a better feasible solution when compared with the other two solvers. Especially the improvement of our BnB algorithm over the CPLEX solver and the BnB-cnt solver is more significant for $(\mathcal{P}_{\text{ccmv}})$, CCMV problem without the no-shorting constraint, than for $(\mathcal{P}_{\text{ccmv}}^{\text{ns}})$, CCMV problem with the no-shorting constraint.

From our tests, we have found that CCMV problems with the no-shorting constraint are relatively easier to solve (by our BnB algorithm) than the general CCMV problems (without the no-shorting constraint). This statement is based on our experience in solving the continuous relaxation for these problems. For problem $(\mathcal{P}_{\text{ccmv}})$, it is rare to observe 0 component in the solution vector from the continuous relaxation. However, for problem $(\mathcal{P}_{\text{ccmv}}^{\text{ns}})$, when the continuous relaxation $(\mathcal{P}_{\text{cnt}}^{\text{ns}})$ is solved, a large percentage of the components in the portfolio are set at 0 in its optimal solution. This fact leads to a conclusion that, when considering CCMV with the no-shorting constraint, only a small portion of the variables are truly free for us to adjust in constructing the portfolio, thus reducing the dimension of the problem under consideration.

Although a straightforward implementation of our BnB algorithm already shows certain advantages over the CPLEX solver and the BnB-cnt algorithm, there is still much room for us to improve our BnB solver. One bottleneck of our BnB solver is the computational burden in calculating bounds $v(\mathcal{P}_{\text{ellp}})$ and $v(\mathcal{P}_{\text{scop}})$. When we solve problems of size $n = 300$, our algorithm also suffers the numerical instability of the SDPA solver in some cases. We believe that if, instead of using general purpose SDP and SOCP solvers, we would use algorithms specifically

designed for taking advantage of the structural properties of the cardinality constrained mean-variance problems, the overall performance and stability could be significantly improved.

7. Conclusion

Stimulated by the long-standing challenge (from both academic and practical viewpoints) of the optimal cardinality constrained portfolio selection problem, we have investigated in this paper efficient exact solution algorithms for a class of such problems. Instead of relaxing the cardinality constraint directly as a majority of the literature has suggested, we consider *separable relaxation* by modifying the primal objective function to some separable functions, which are immune to the hard cardinality constraint. We believe that keeping the cardinality constraint has an advantage as it preserves certain inherent features of the primal problem, thus enabling us to exploit rich geometric properties of the problem. A combination of an analytical solvability of the cardinality constrained separable relaxations and their corresponding polynomial-time dual search algorithms empowers our BnB algorithm with a capability to offer tight bounds and good feasible solutions with relatively cheap computational costs, and to outperform the CPLEX solver significantly for problems with a relatively large dimension.

The proposed solution scheme in this study may have potential to be applied to other hard problems in a wide spectrum of OR applications, e.g., the facility location problem with the convex quadratic transportation cost (see Günlük et al. 2007), where cardinality is a major concern.

Supplemental Material

Supplemental material to this paper is available at <http://dx.doi.org/10.1287/opre.2013.1170>.

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